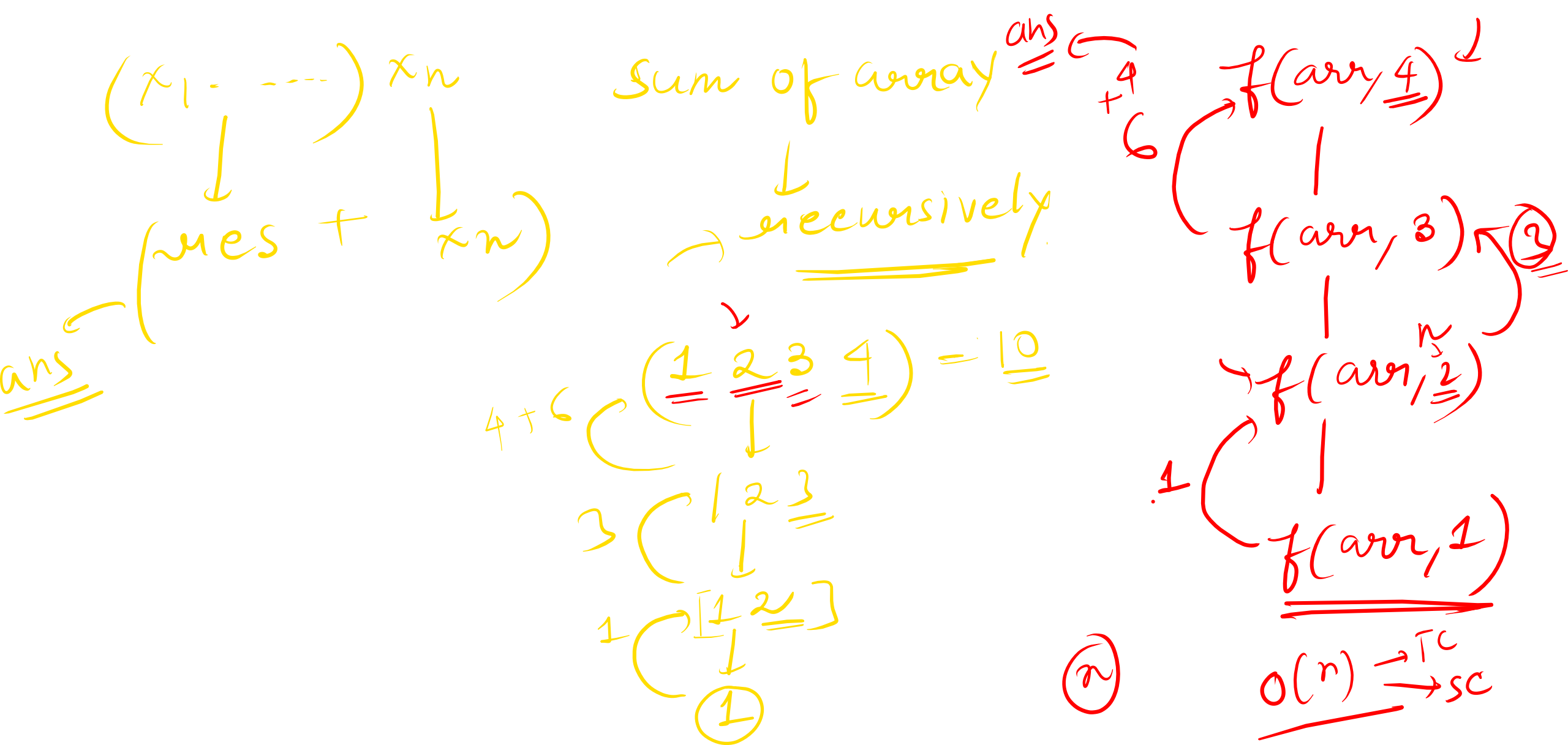
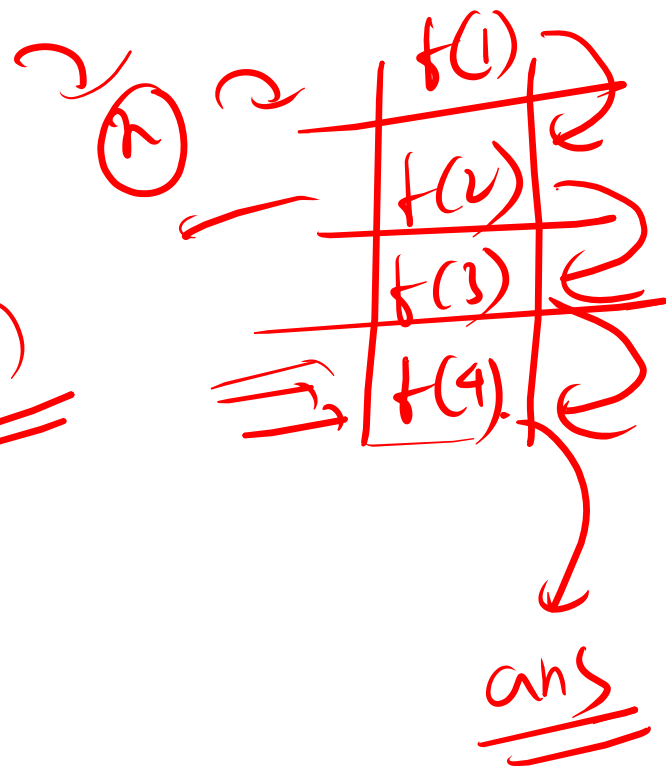




$O(n)$



$f(arr, n) \{$

$c_1 \leftarrow \{$   
if  $(n == 1)$   
return  $arr[0];$   
 $\underline{res} = f(arr, n-1);$   
return  $\underline{res + arr[n-1]}$ ;  
 $\}$

$n * c$   
 $O(n)$   $\rightarrow$  TC

$$O(n \cdot \phi)$$
 $O(n) \rightarrow TC$ 

$O(n+1) \rightarrow SC$

 $O(n)$ 

Linear search  $\rightarrow$  Recursively

①

$\{K=4\}$

$$\left( \begin{array}{cccc} \textcircled{1} & 5 & 2 & 3 \\ & & & \end{array} \right) \xrightarrow{\quad} \underline{\underline{n=4}}$$
$$(1 \ 5 \ 2) \rightarrow \eta = 3$$

15

$$(1, 5) \rightarrow n=2$$

$\psi_n = 0$   
 $\psi_n \in$   
 $\uparrow$

$$(1)_{n=1}$$
$$\rightarrow (1, 5) \rightarrow n=2$$

$f(\text{over}, n)$

$$\otimes \int \underline{\underline{n-1}} \rightarrow K$$
$$f(arr, n-1)$$

$f(\text{array}, k, n)$

if ( $n == 0$ )  
return -1;

if ( $\text{array}[n-1] == k$ )

return  $n-1$ ;

return  $f(\text{array}, k, n-1)$ ;

Tail  
recursion  
?

$k = \textcircled{2}$

1 2  $\textcircled{3}$  4

$f(4) \rightarrow \text{main}$   
 $n = 4$

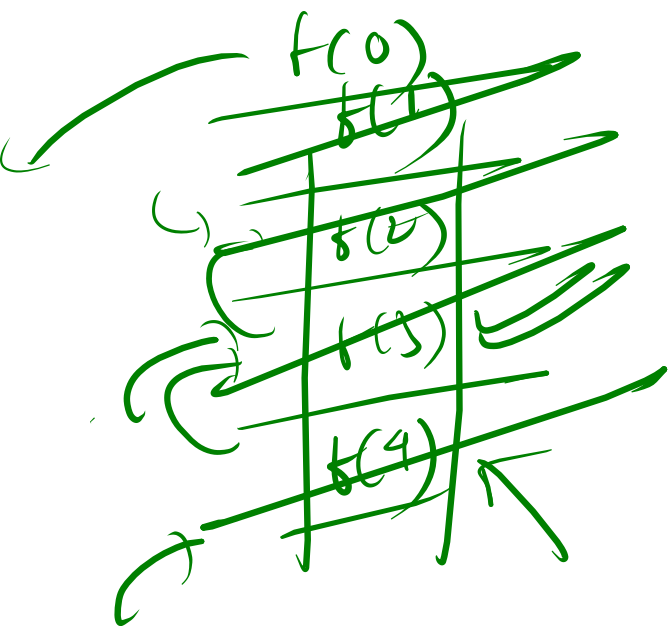
$f(3) \rightarrow n = 3$

$f(2) \rightarrow n = 2$

$\textcircled{1}$

IC/SC





$\left\{ \begin{array}{l} \text{mes} = f(n-1); \\ \text{return mes}; \end{array} \right\} \otimes$

$\xrightarrow{\text{return}} \boxed{f(n-1)};$   
 $\otimes \rightarrow +, *, /, *$

compiler  
↓

Java  $\otimes$

$\frac{\text{C++} \checkmark}{\text{C} \checkmark}$

3  $\otimes$   
 2  $\otimes$   
 1  $\otimes$   
 0  $\otimes$

{arr, k}  
↓  
class

$f(n)$  { main dependency

if (n == 0)  
return n - 1;

if (arr[n-1] == k)  
return n-1;

parent call

return  $f(n-1)$ ;  
}



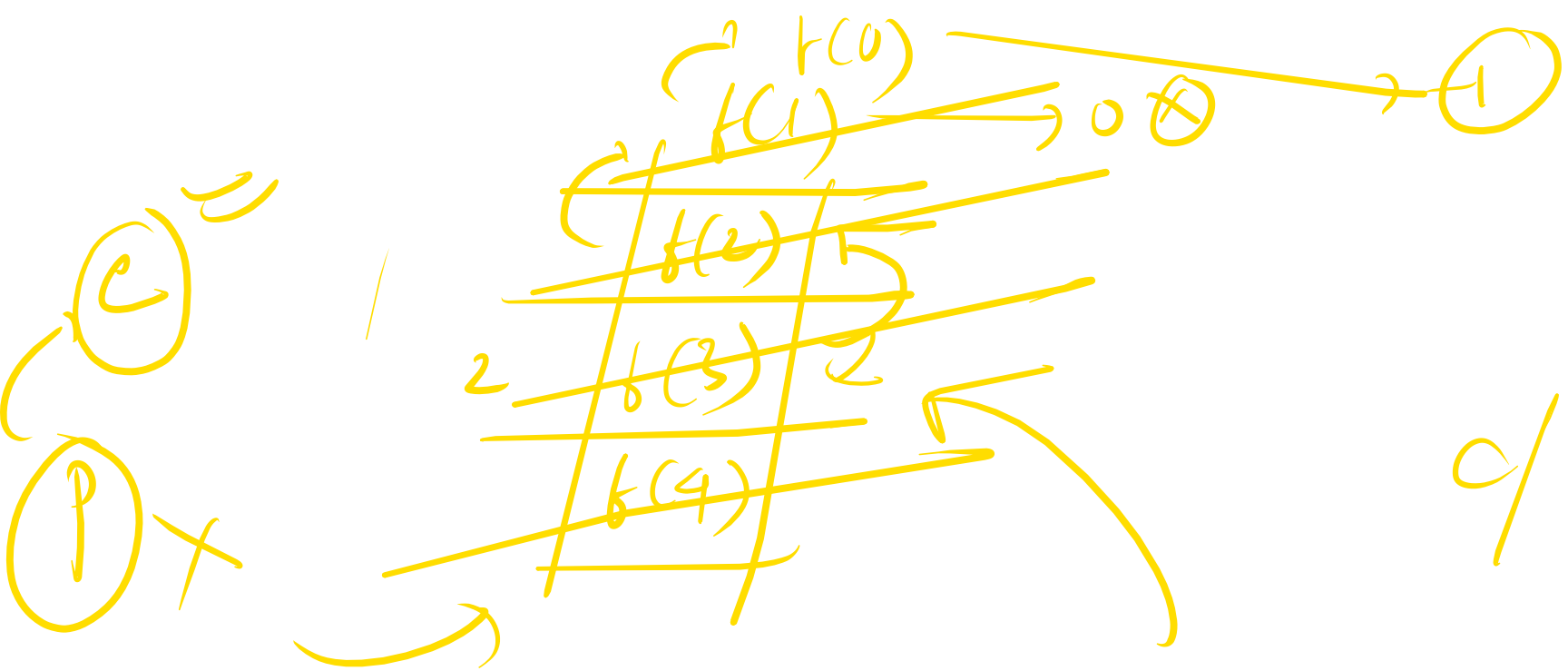
⊗  $O(n)$

curr function

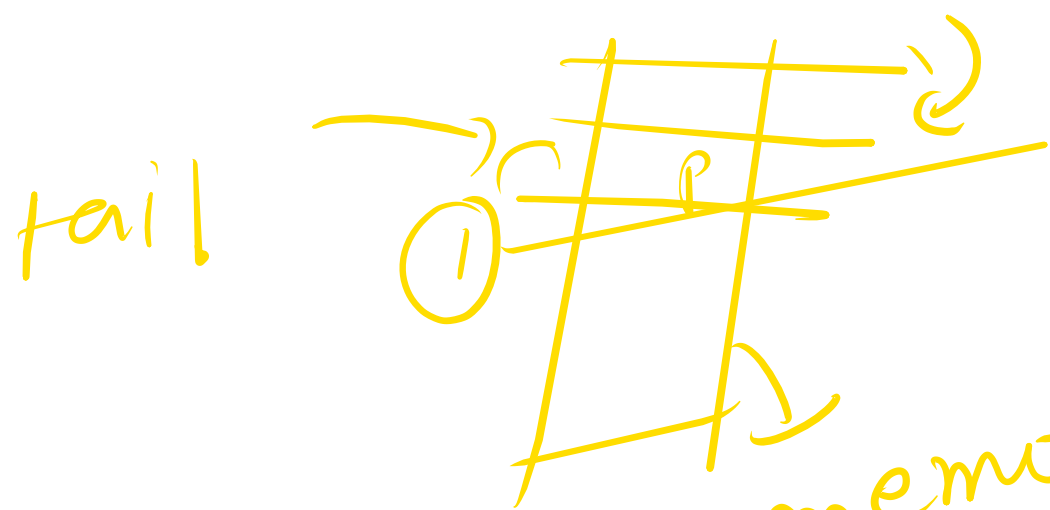
sum → return  $f(n-1) + arr[n-1]$ ;  
fact → return  $n * f(n-1)$ ;

non tail recursive.

tail recursion



c/c++ ✓



memory optimization

Java

gcd  $a/b \rightarrow$  recursively

euclid's  
theorem  
?

$$\gcd(\bar{a}, \bar{b}) \equiv \gcd(b, a/b)$$

main

2

ged(10, 4)



ged(4, 2)



ged(2, 0)



a



b == 0

f(a, b) {

if (b == 0)

return a;

return

ged(b, a/b)

;

Tail

recursion



$k = \textcircled{1}$

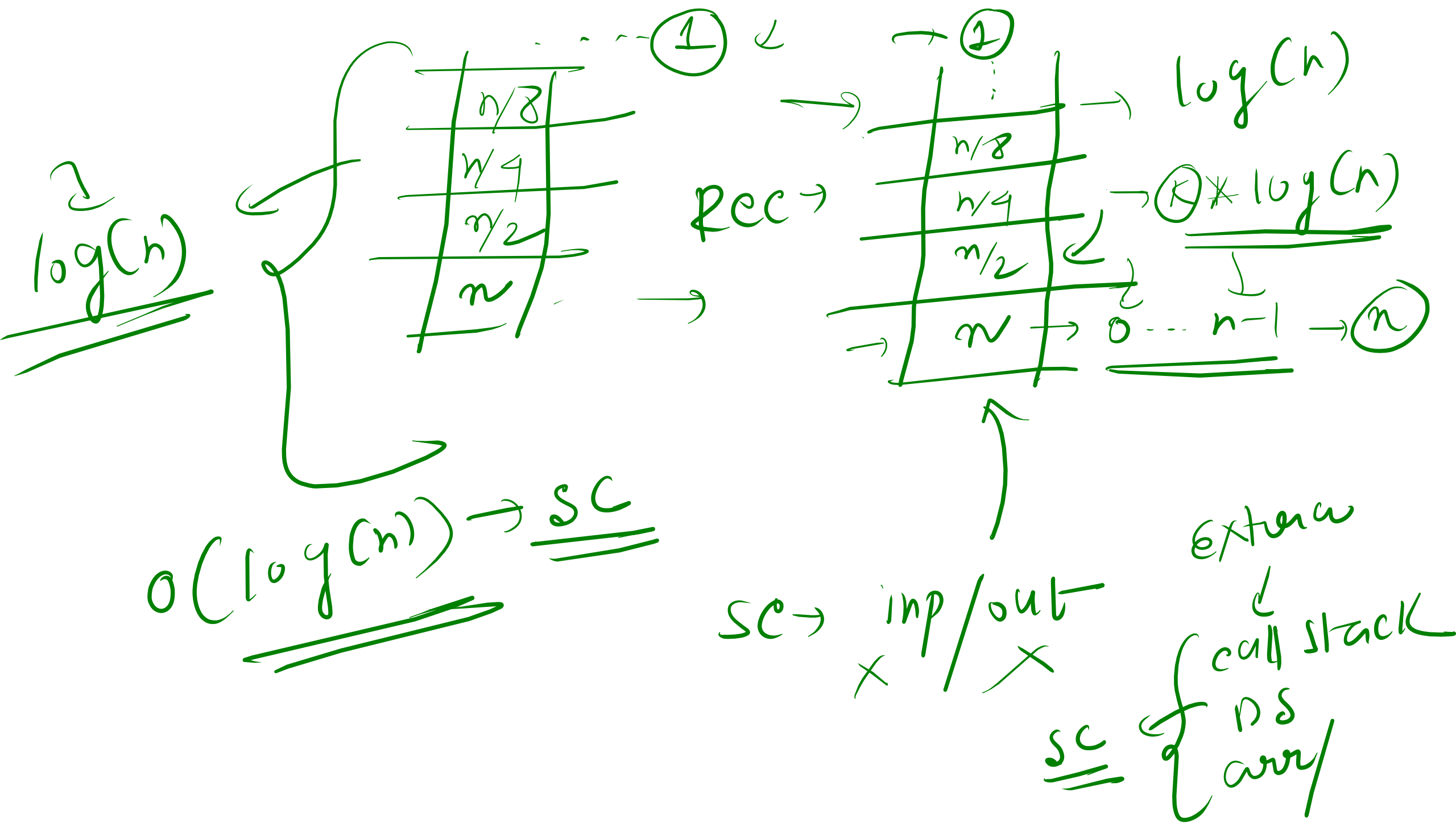
arr Binary search  
↓  
Recursion

1 2 4 5 9

→  $f^2(0, 4) \rightarrow \frac{2}{4} > 1$   
0 (↓  
 $f(0, 1) \rightarrow \underline{\underline{0}}$

```
f(st, en) {  
    if (st > en) ↵  
        return -1;  
  
    mid = (st + en) / 2;  
    if (arr[mid] == k)  
        return mid;  
}
```

```
    if (arr[mid] > k)  
        return f(st, mid-1);  
    return f(mid+1, en);  
}  
↵  $O(\log n)$ 
```





# Extended euclid

↓

$$5 \div 3 = 2$$

$$5 - (3) = 2$$

$$5/3 = (1) * 3 \rightarrow 2 \quad \boxed{5} \quad \text{remainder}$$

multiplicative  
modular  
inverse

$$\overset{2}{a}x + \overset{1}{b}y = \text{gcd}(a, b)$$

$$= \text{gcd}(b, a/b) =$$

$$= bx_1 + (a/b)y_1$$

$$= bx_1 + (a \lfloor a/b \rfloor * b)y_1$$

$$\underline{ax + by} = bx_1 + ay_1 - \lfloor a/b \rfloor by_1$$

$$\underline{ax + by} = \underline{b}(x_1 - \lfloor a/b \rfloor y_1) + \underline{ay_1}$$

ans

$$\left\{ \begin{array}{l} \textcircled{x} = y_1 \\ y = x_1 - \lfloor a/b \rfloor y_1 \end{array} \right\}$$

~~$x \neq 0$~~   
 $x = 1 \rightarrow 1$   
 $y = 0 - 2 * 1 \rightarrow -2$

$10, 4 \rightarrow \underline{x, y} \leftarrow (0, 1)$   
 $\downarrow$   
 $4, 2 \rightarrow \underline{x_1, y_1} \leftarrow (1, 0) \rightarrow x_1 = 0?$   
 $y_1 = 1 - 2 * 0$

$\underline{x, y_1}$   
 $\downarrow$   
 $2, 0 \rightarrow \underline{x_2, y_2}$   
 $\text{②}$

$ax + by = \boxed{a}$   
 $\downarrow$

$\left( \begin{matrix} ax \\ \downarrow \\ 1 \end{matrix} + \begin{matrix} 0y \\ \downarrow \\ 0 \end{matrix} = a \right)$

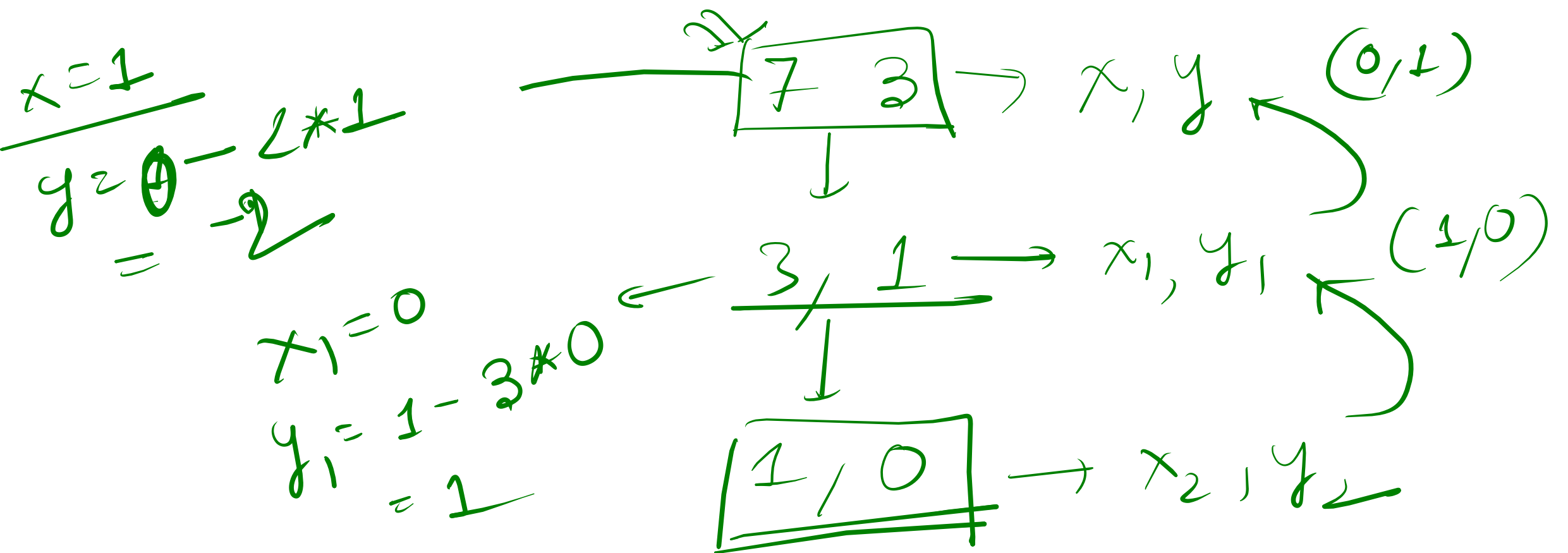
~~$x, y$~~   
 $x = y_1$   
 $y = x_1 - [a/b]y_1$

multiplicative  
modular  
inverse.

$$\left\{ \frac{10, 4}{\underline{\quad}} \right\} \rightarrow \frac{x = \textcircled{1}}{\underline{y = -2}} \rightarrow \underline{\underline{MMI}}$$

$$10x + 4y = \underline{\underline{2}}$$

$$10 * 1 + 4 * (-2) = 2 ?$$



① MMI

$$7 * (1) + (3) * (-2) = \underline{\underline{1}}$$

$$x \rightarrow \underline{MMI} ?$$

$$ax + by = 1$$

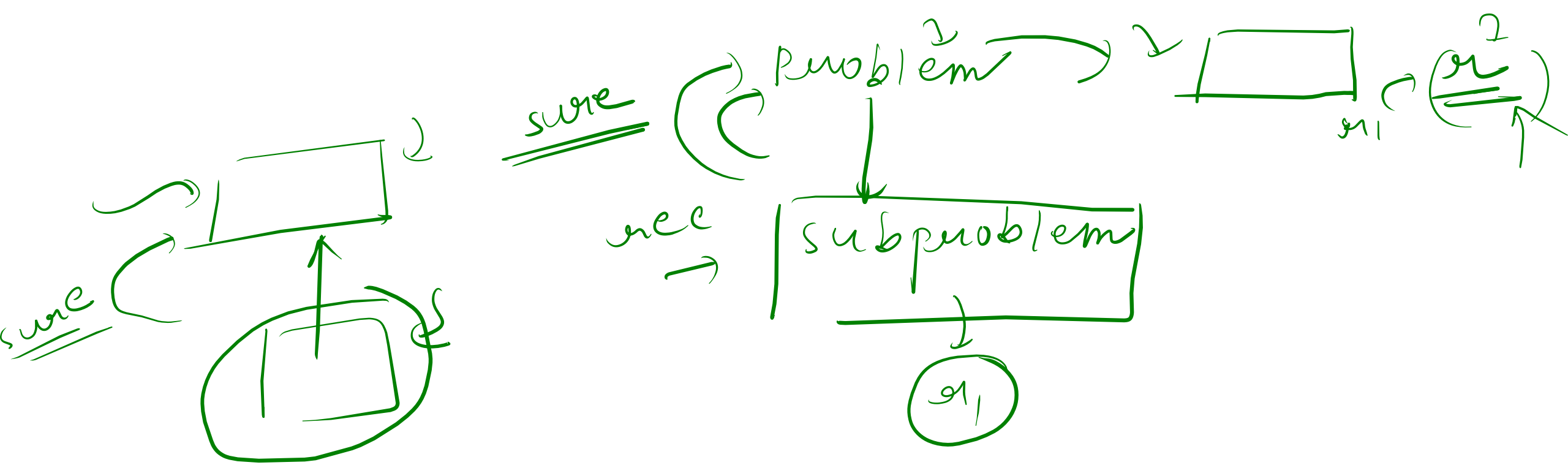
$$\left( by \cdot b \right) = \underline{\underline{0}}$$

$$\left( \overset{2}{a^{-1}} \cdot b \right)$$

$$\left( \frac{ax}{\cancel{b}} + \frac{by}{\cancel{b}} \right) \cdot b = \left( 1 \cdot b \right)$$

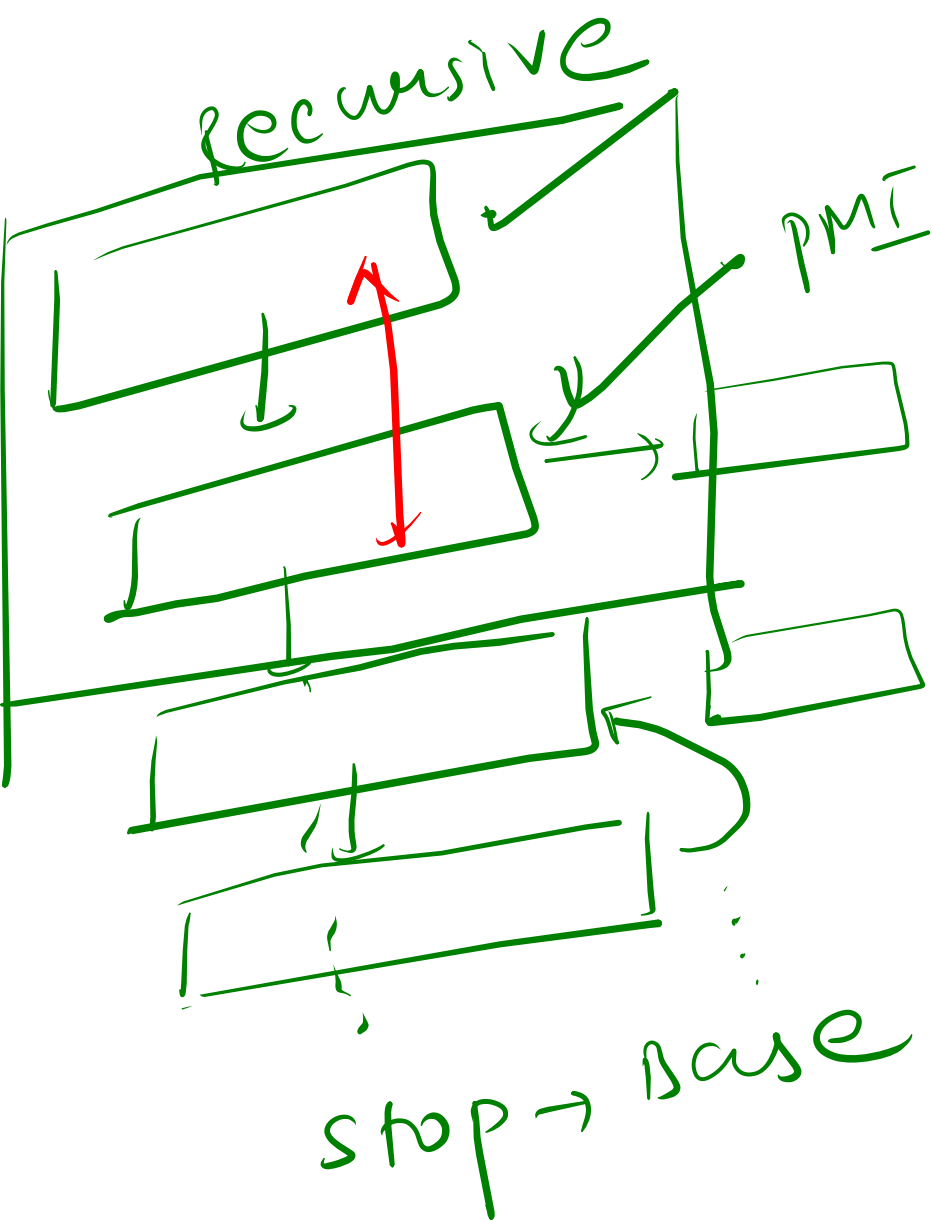
$$(ax) \cdot b = (1 \cdot b)$$

$$\underline{\underline{a^{-1}a}} \cdot \left( \overset{2}{x} \cdot b \right) = (a^{-1}) \cdot b$$



0/1 ✓ PMI → 11 → Recursion

→  $K^2 + K + 1$  → xyz



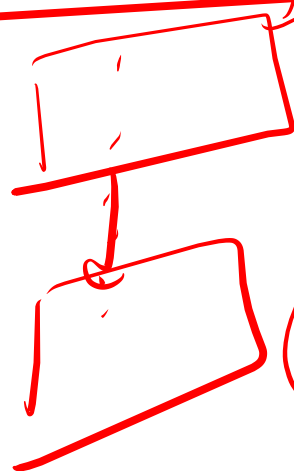
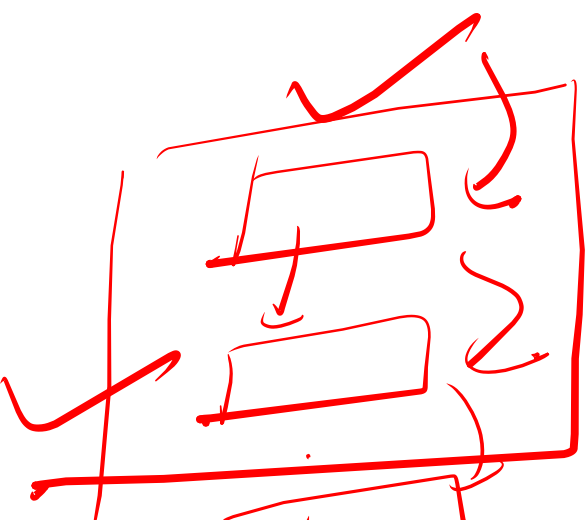
Base

$x/y/z \rightarrow$  pre defined nos.

sub  $\rightarrow$   $n_1$

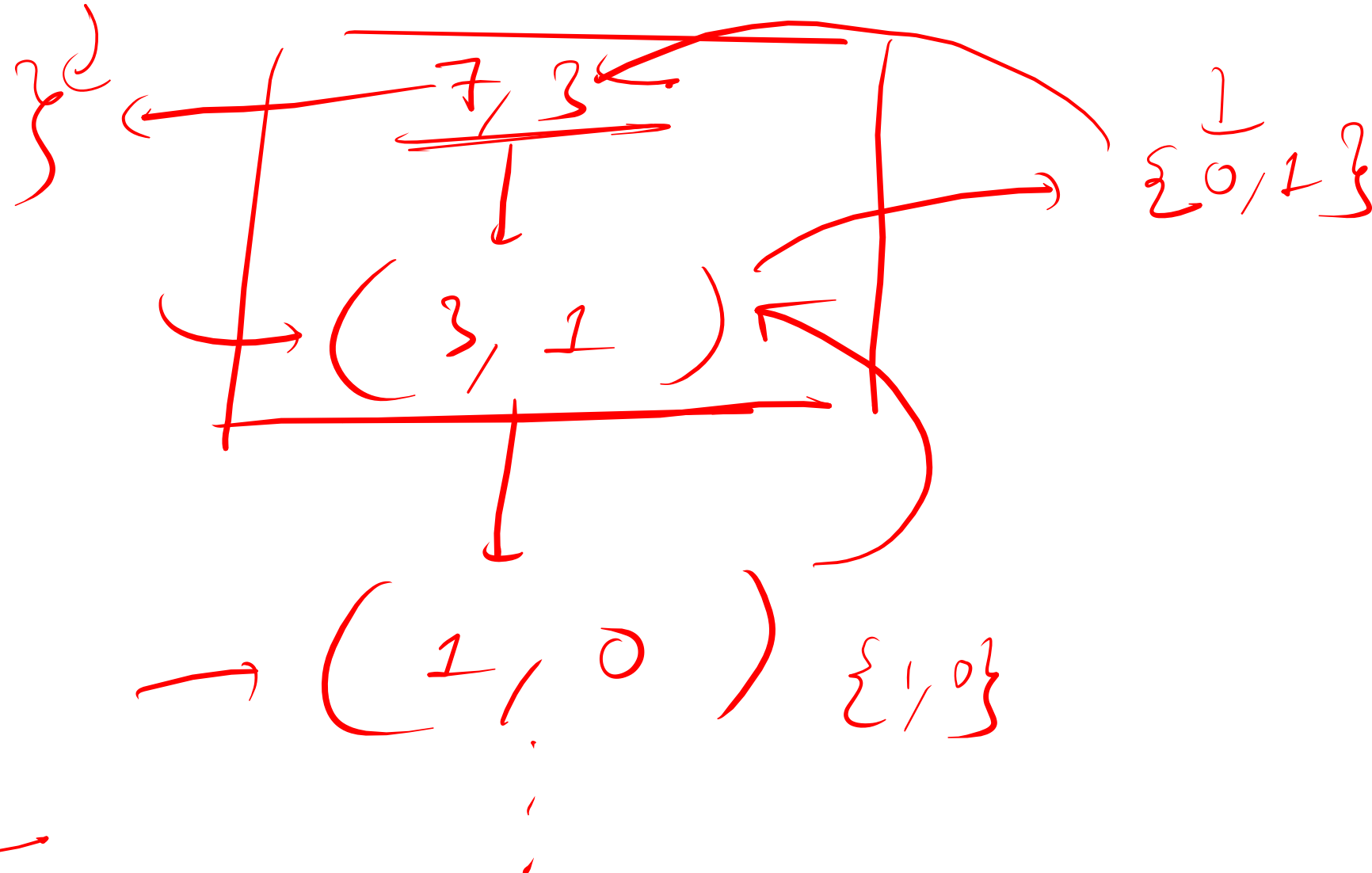
$n_1$   $\rightarrow$   $(n)^2$





22  
 (8) 5

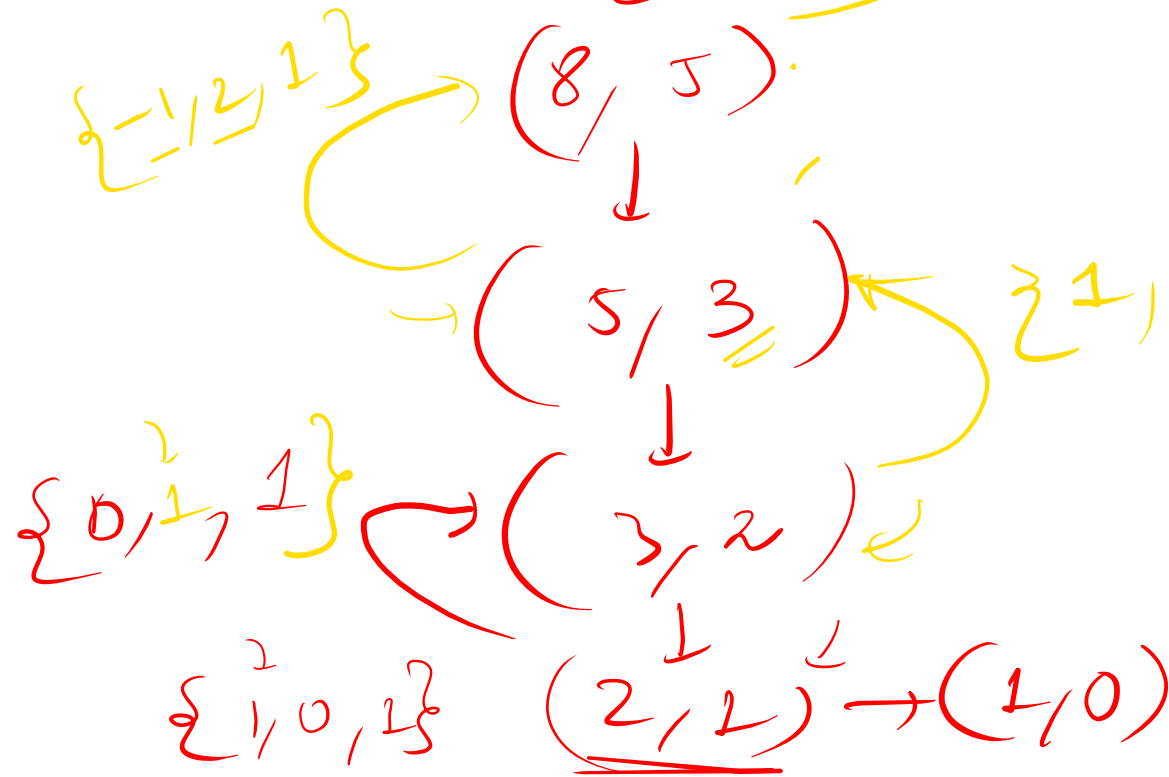
a > b





$$x = a_1$$

$$y = x_1 - a/b a_1$$



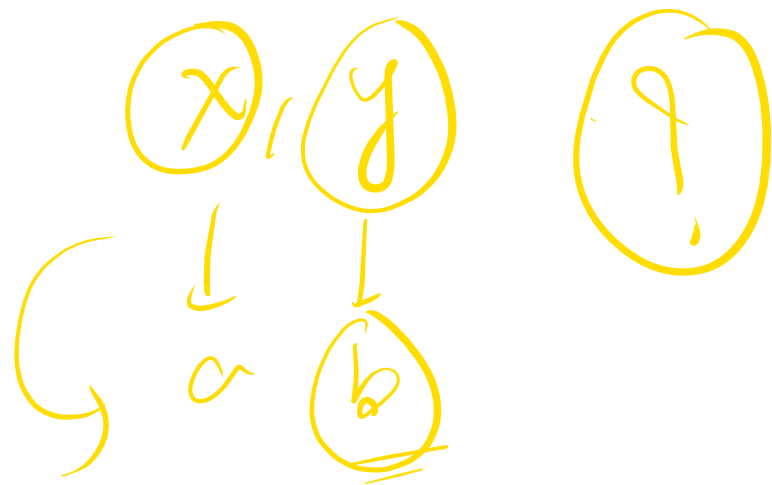
$$5 \times (-5) + 8 \times 2 = 1$$

$$25 + 16 = 1$$

$$(41) : 8 = 5$$

$$5 \times (-3) + 8 \times 2 = 1$$

$$(-3 + 8) : 8 = 5?$$



$$\underline{\underline{8x + 5y = 1}}$$

↓

$$\underline{\underline{x = 2 \quad y = -3}}$$

↪

$$\underline{\underline{a > c}}$$

$$\textcircled{x} / y \rightarrow -ve$$

-ve  
↪

$$\textcircled{(x + b) / b}$$

↙ = ?

$$\begin{array}{c} \downarrow \\ \textcircled{n} \rightarrow \text{odd} \\ \downarrow \\ +1 + (n/2) * 2 \end{array}$$

$$\begin{array}{c} x^n \\ \downarrow \\ (x^{n/2})^2 \end{array}$$

$$\begin{array}{c} x^n \rightarrow \underline{\underline{\log(n)}} \downarrow \\ \underline{\underline{recursively}} \end{array}$$

$$\begin{array}{c} n=5 \\ (x^2 * x^2) = \underline{\underline{x^4 * x}} \\ \underline{\underline{x^5}}? \end{array}$$

$$8 \times 8 \leftarrow (2, 6) \rightarrow n=6$$

↓ ↗<sub>8</sub>

$$2 \times 2 \times 2 \leftarrow (2, 3) \rightarrow n=3$$

↓ ↗<sub>2</sub>

$$1 \times 1 \times 2 \leftarrow (2, 1) \rightarrow \underline{n=1}$$

↓ ↗<sub>1</sub>

$$(2, 0) \rightarrow n=0$$

$$\begin{array}{c} n \\ | \\ n/2 \\ | \\ n/4 - n/8 \dots 0 \end{array}$$

$\log(n)$

int pow(x, n) {

if (n == 0)

return 1;

---

int res = pow(x, n/2);

---

if (n % 2 == 1)

return res \* res \* x;

return res \* res;

}



abc  
bdc  
efg

$$\begin{array}{r} aa \\ \hline aab \\ \hline aac \end{array} \quad \left. \vphantom{\begin{array}{r} aa \\ aab \\ aac \end{array}} \right\}$$
$$\underbrace{\quad \quad \quad}_{\text{②}} \quad \quad \quad \underbrace{\quad \quad \quad}_S^h$$

# Terie

# Retrieval

extracted

$\delta$   $\rightarrow$  rate  $\rightarrow$   $\text{len}(s)$

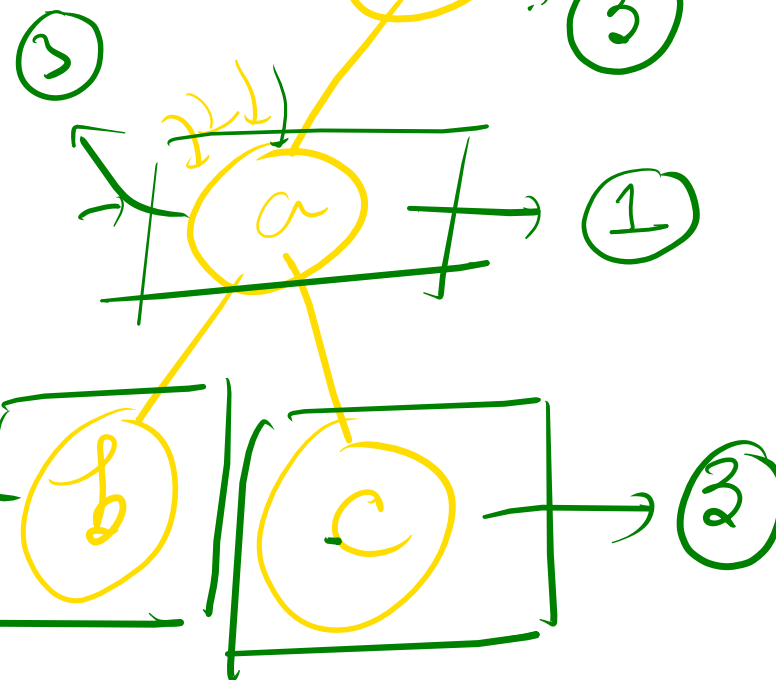
$\frac{2}{aa}$ 
 $\frac{2}{aue}$   
 $\frac{2}{aab}$ 
 $\frac{2}{C}$

# Root

B  
nodes

4

sch



BIT trie

0/1

7 5 3

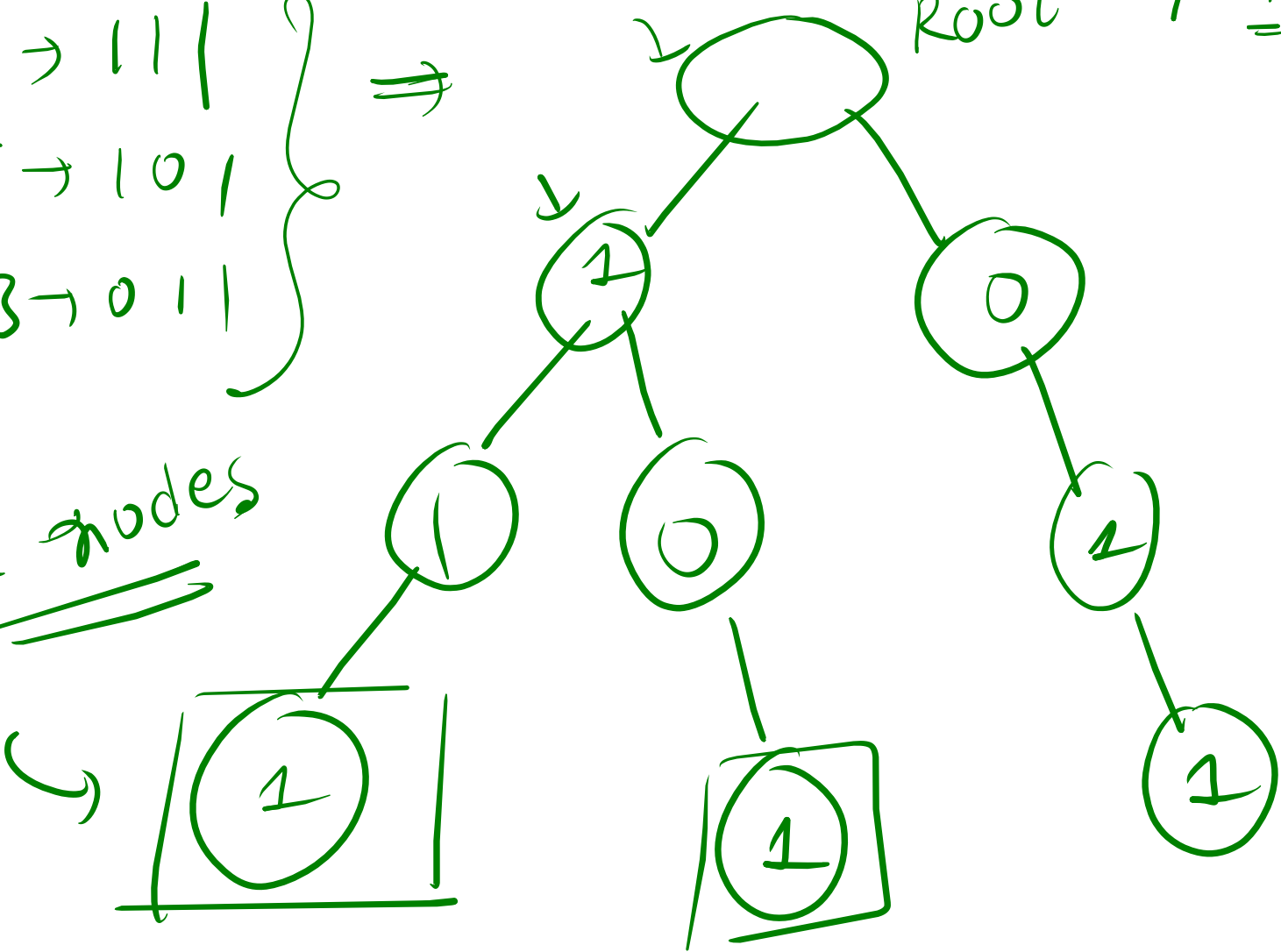
int → { 32 bit }

{  
7 → 111  
5 → 101  
3 → 011  
}

⇒

Root 7 - 111

32 nodes



```
insert(Node * root, string s) {
```

Node ?

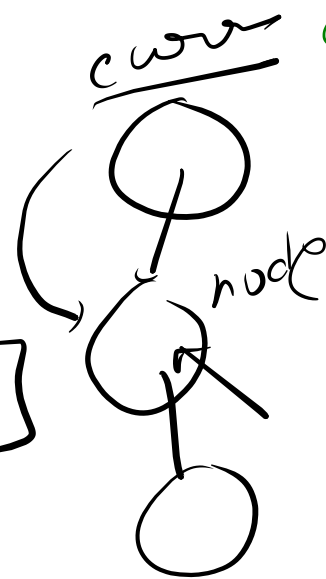
```
Node * cur = root;
```

```
for(char ch : s) {
```

```
if(cur->arr[ch-'a']  
== NULL) {
```

```
cur->arr[ch-'a']  
= new Node();
```

```
}
```



```
class Node {
```

```
public:
```

```
Node * arr[26];
```

```
int count = 0;
```

```
bool end = false;
```

```
Node() {
```

```
};
```





```
curr = curr->arr[ch - 'a'];  
curr->count++;
```

```
}
```

```
curr->end = true;
```

```
}
```

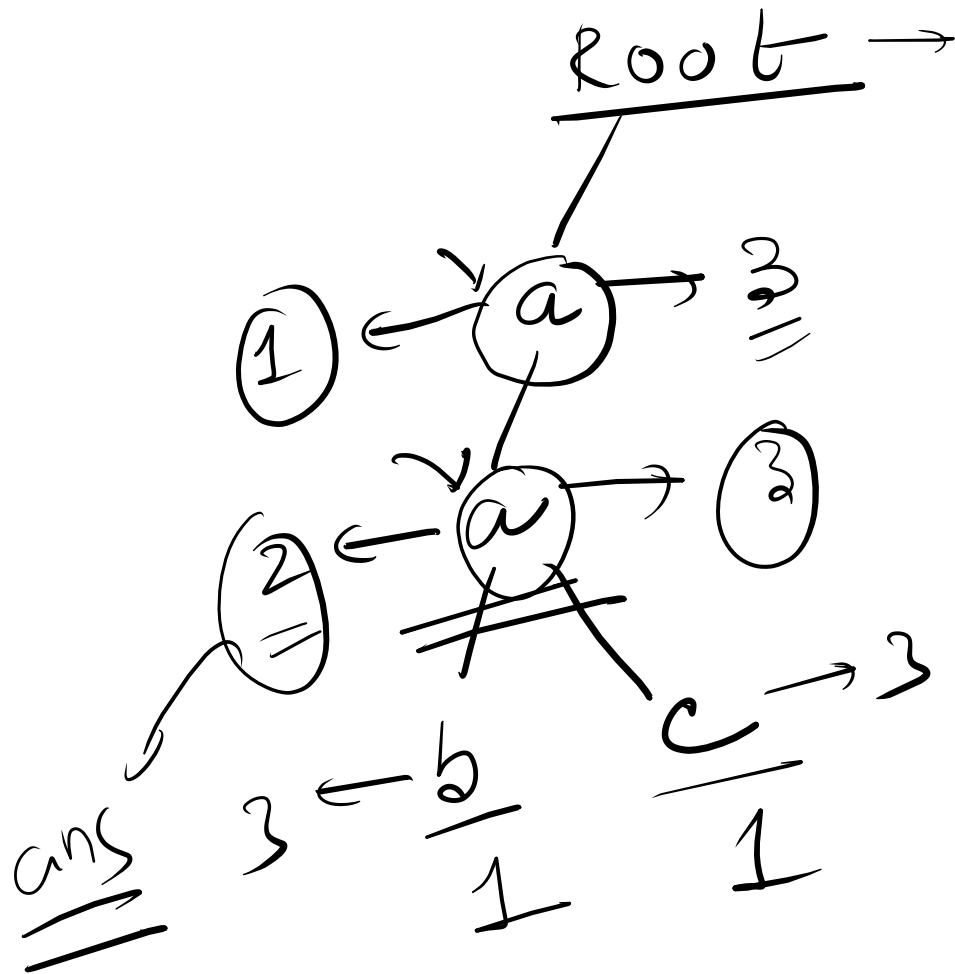
trie  
suffix  
automata

```
bool search(Node *root, string s) {  
    Node *curr = root;  
    for(char ch: s) {  
        if (curr->arr[ch-'a'] == NULL)  
            return false;  
        curr = curr->arr[ch-'a'];  
    }  
    return  
    curr->end;  
}
```

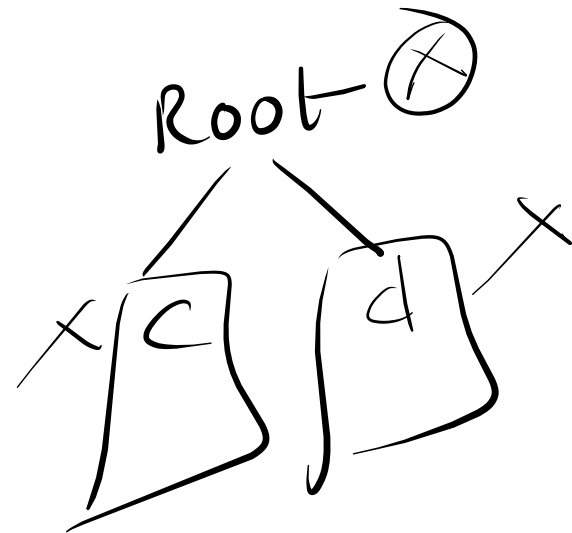
{ n string } longest common prefix.

aa  
2

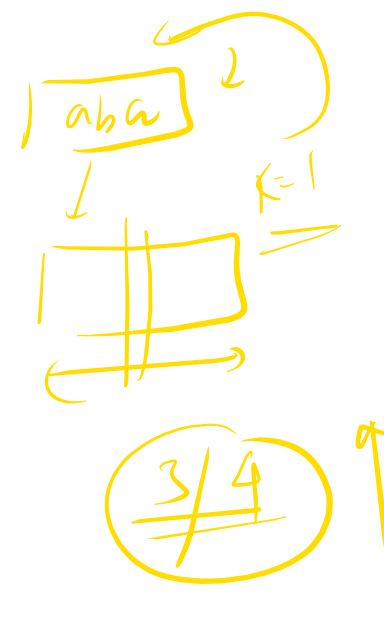
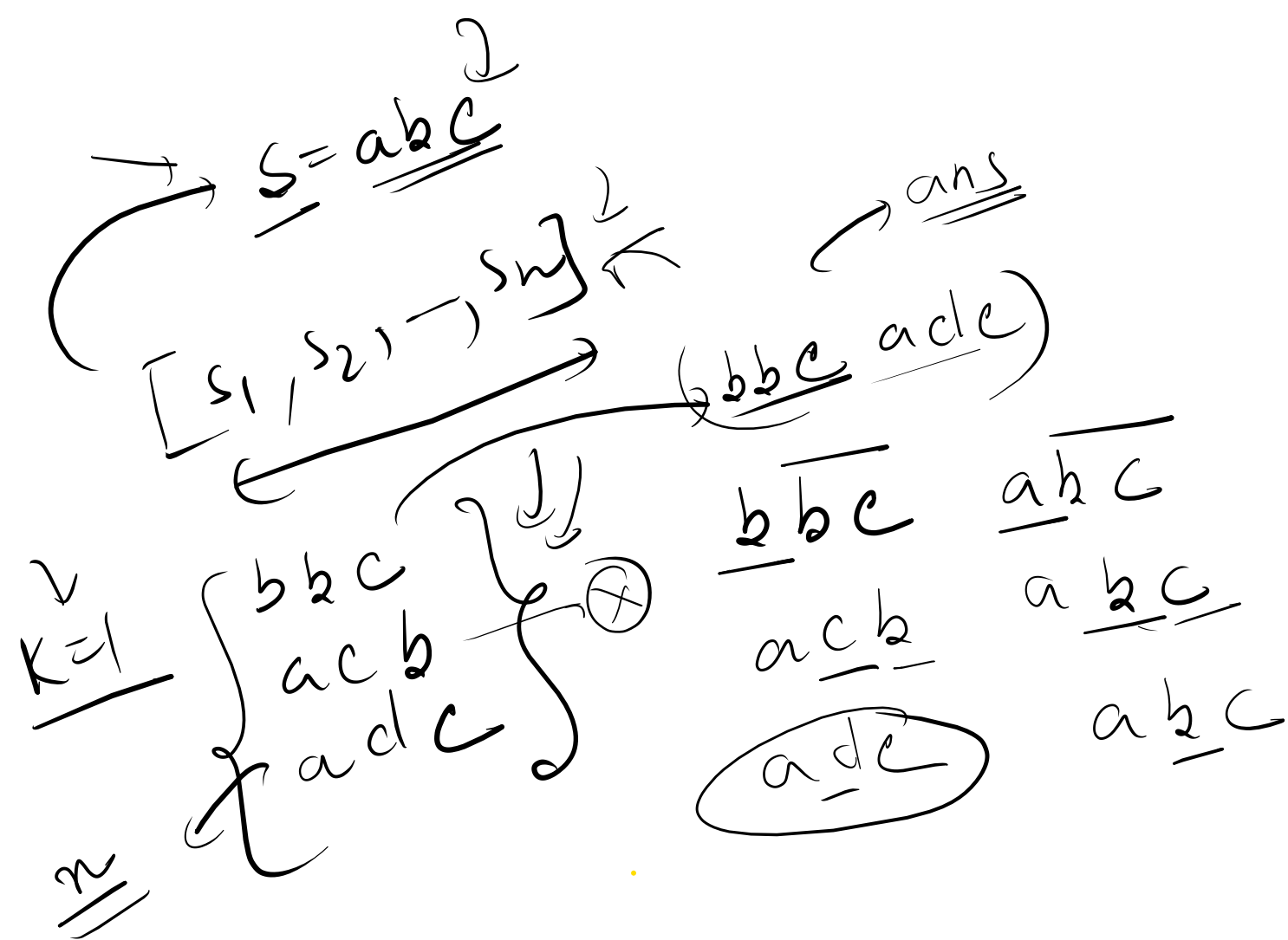
aa  
aab  
aac



abc  
bcc  
cde  
0

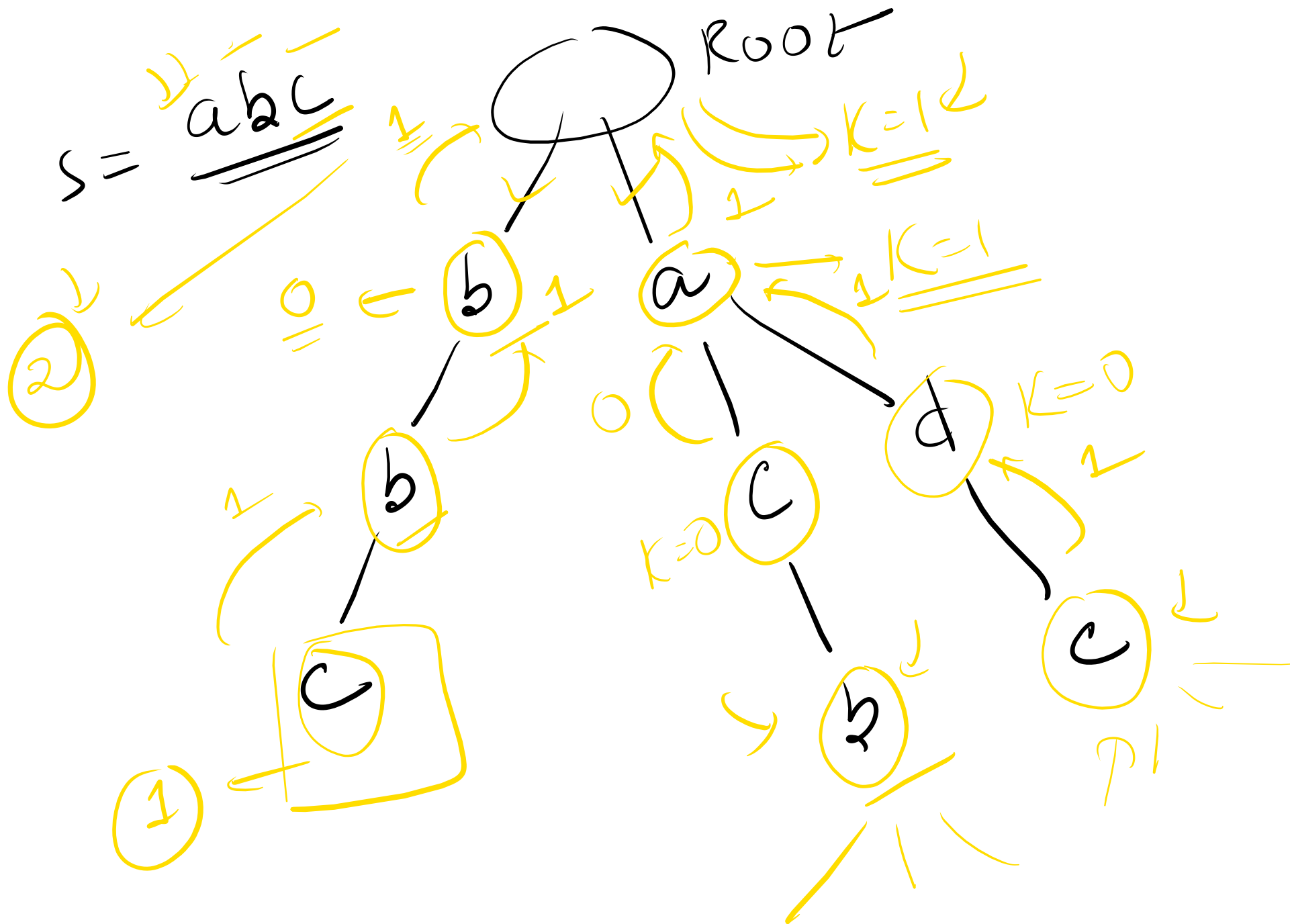


spelling  
mistake  
check



$s = \underline{\underline{abc}}$

max function



root K > 0

2K 1  
1  
1



K > 0

K = ①

$f(\text{Node} * \text{root}, s, K, \text{idx})$

if (idx == len(s))

return 1;

count = 0;

for (i = 0; i < 26; i++) {

if (root->arr[i] != NULL) {

if (s[idx] - 'a' == i && K > 0)

count += f(root->arr[i], s, K-1, idx+1);

}

else {

count += f(root->arr[i], s, K, idx+1);

}

return count;

}

# BIT Trie

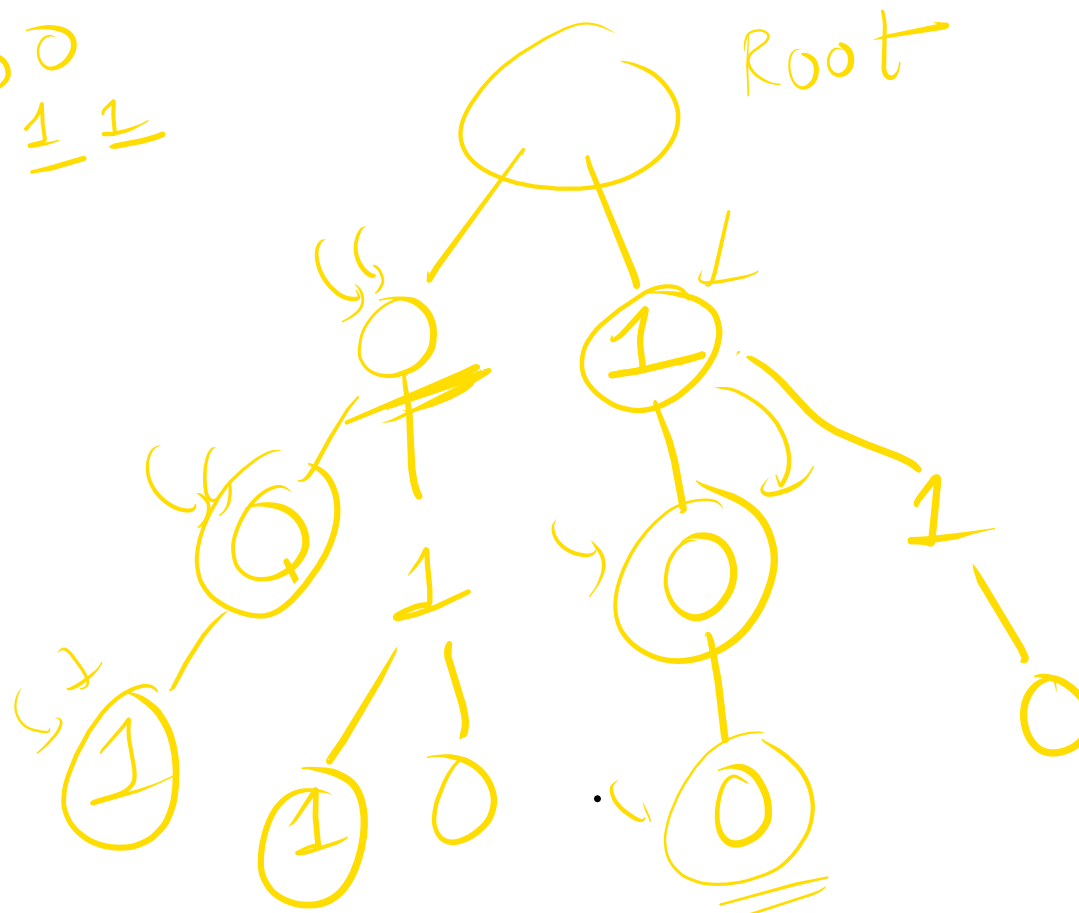
5 → 0

pair  
arr → 1, 9, 2, 6, 3

3  
100  
011

max xor  
value of  
a pair

x14  
xor  
max



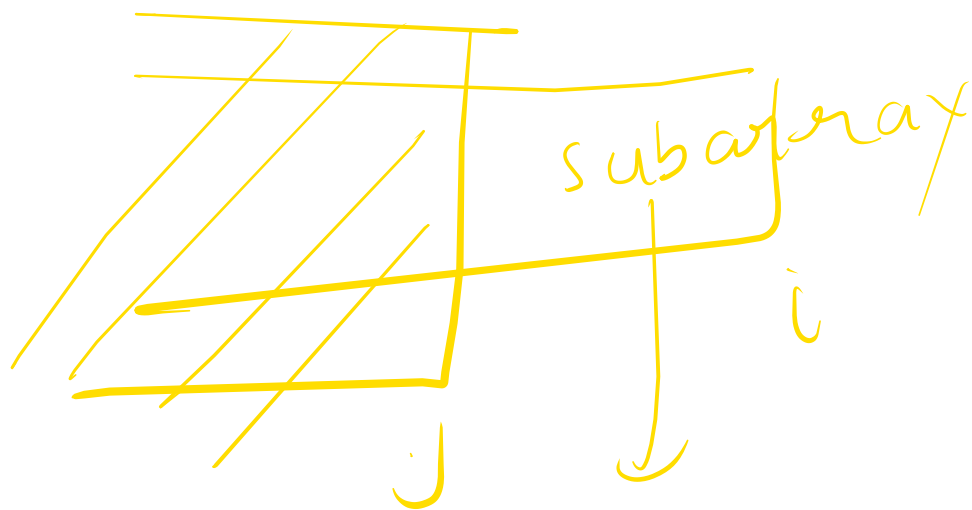
011  
100  
---  
111

ans  
7

TC →  $[n * (32)]$

$O(n)$





xor  $\rightarrow$  max  
ans



BIT Trie ?



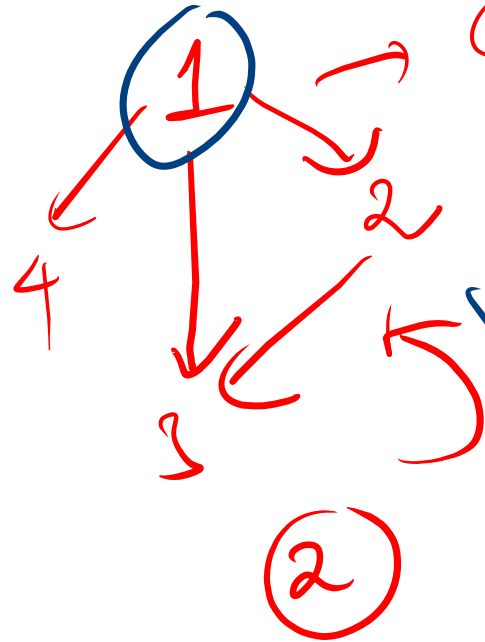
# Cycle detection

DFS

2



cyclic graph

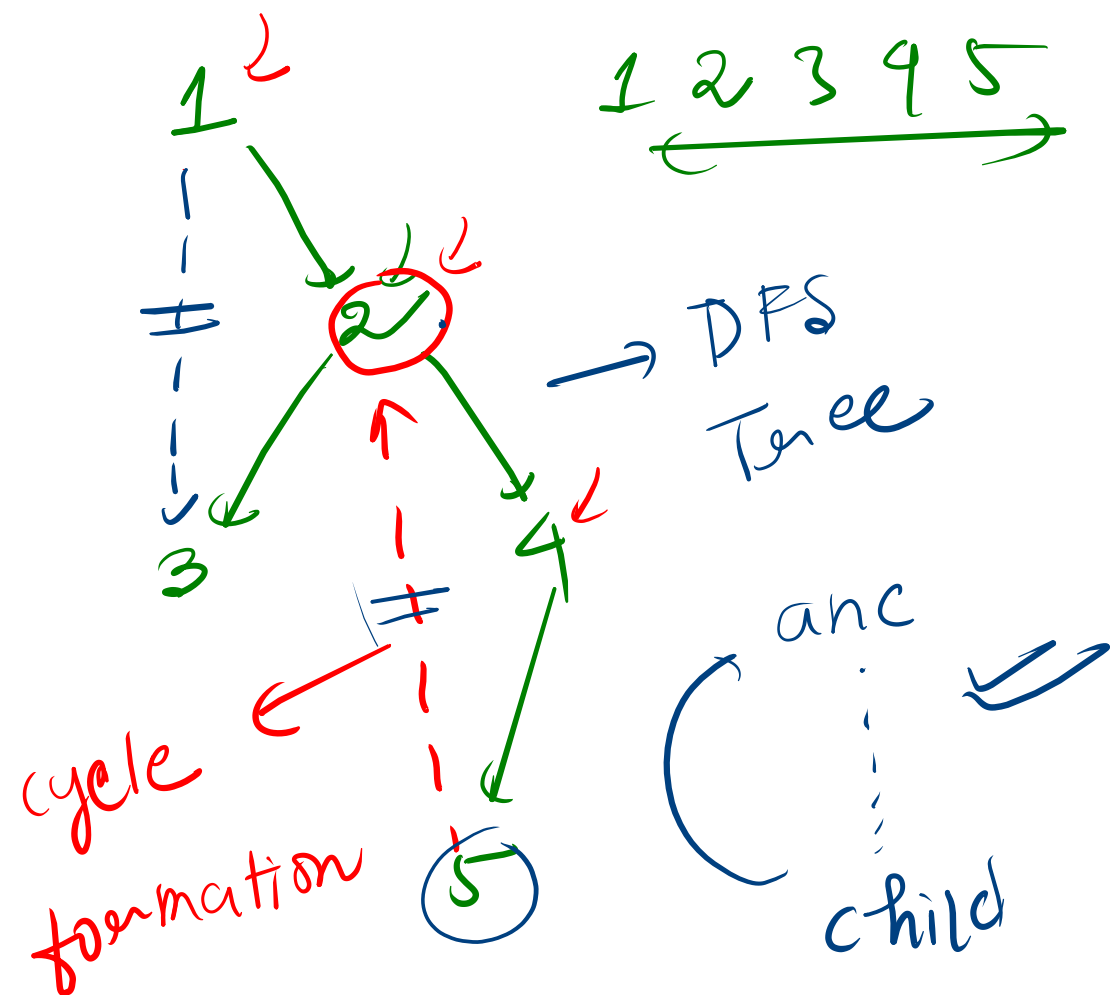
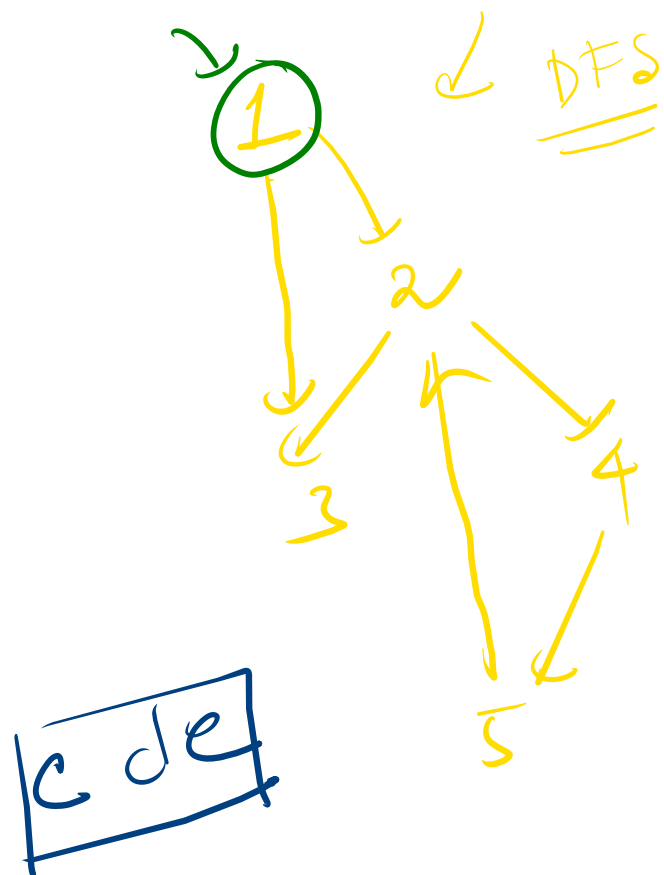
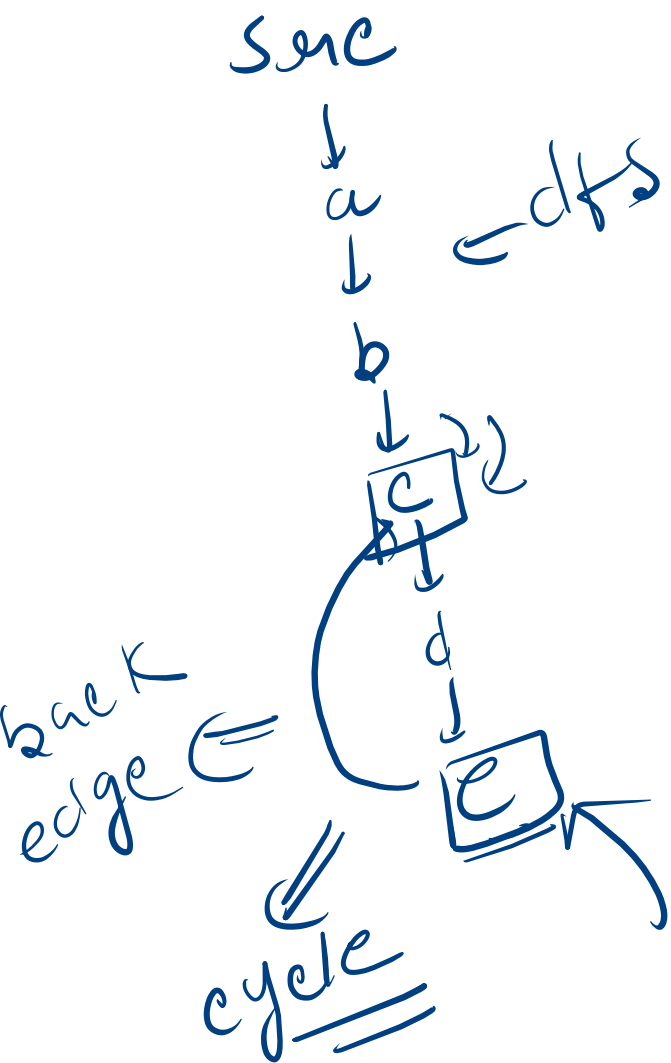


DAG

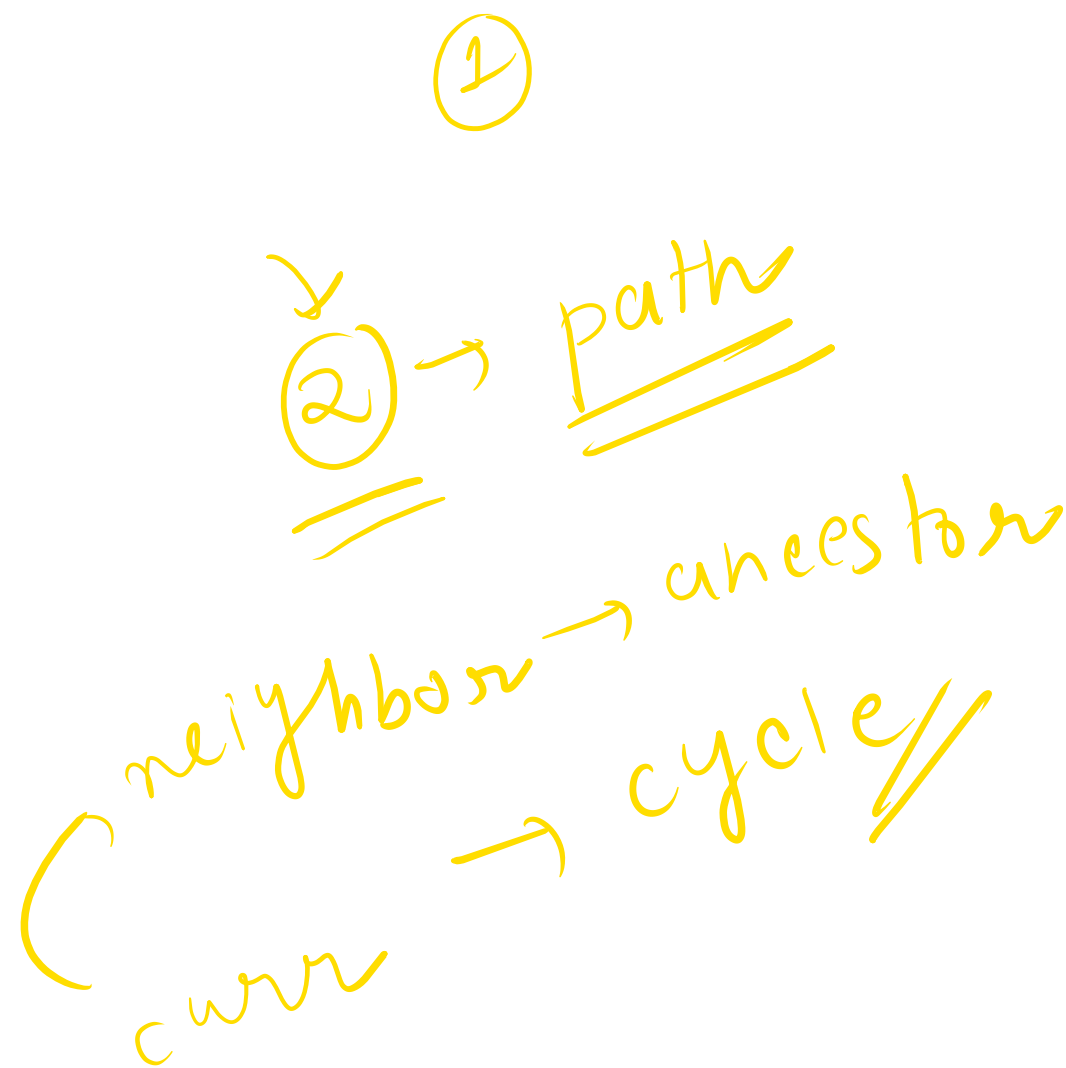
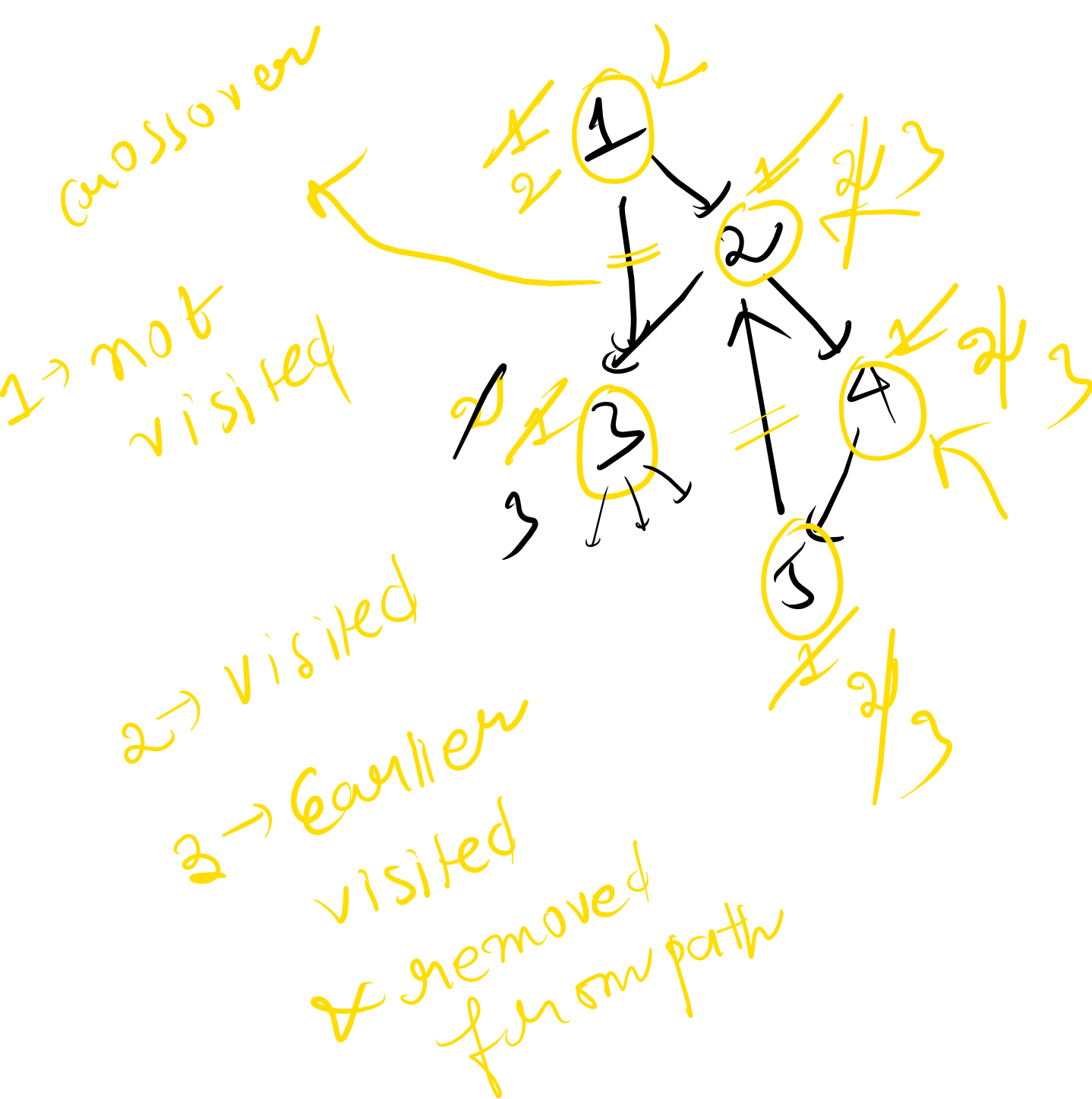
Directed acyclic graph

Topological ordering  
↳ Kahn's algo









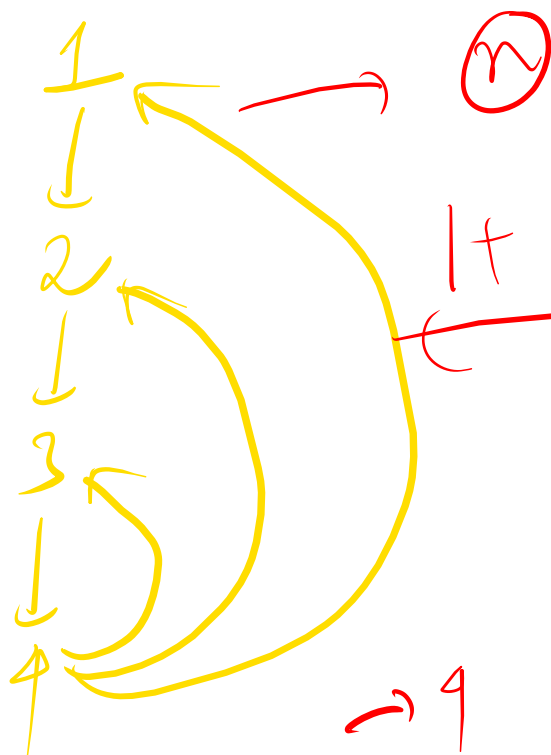
can we print all cycles?

yes

$10^5$  node

$2 \times 10^5$  edges.

dfs



$$1 + 2 + 3 + 4 + \dots + n$$

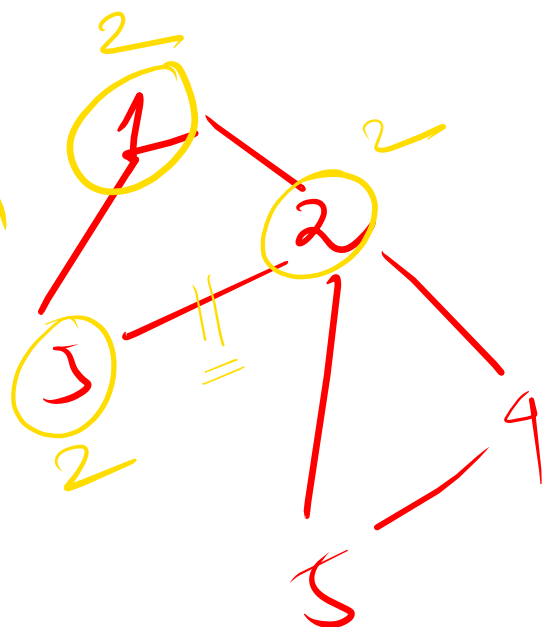
$$\underline{\underline{O(n^2)}}$$



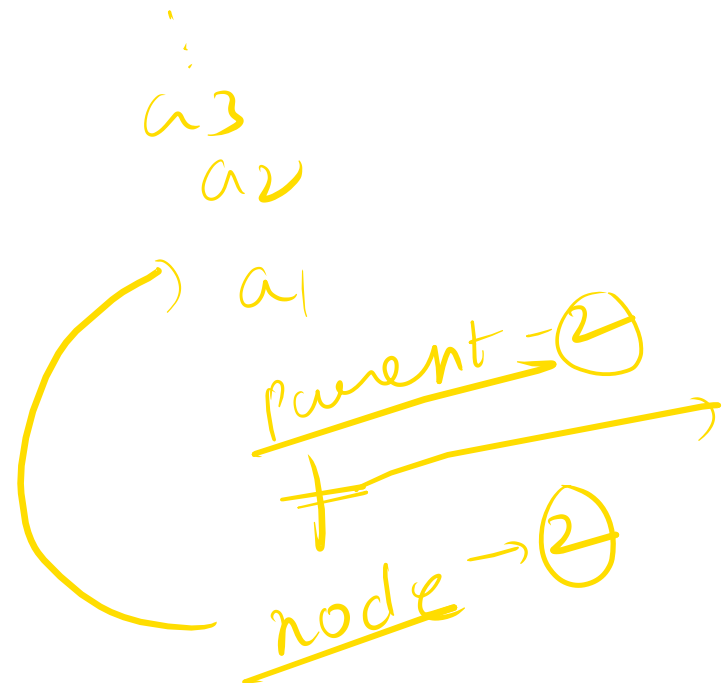
3  
2  
1



returns  
true. ↗  
?



parent  
≠ → back  
node edge



node → parent  
↓ ⊗  
back edge

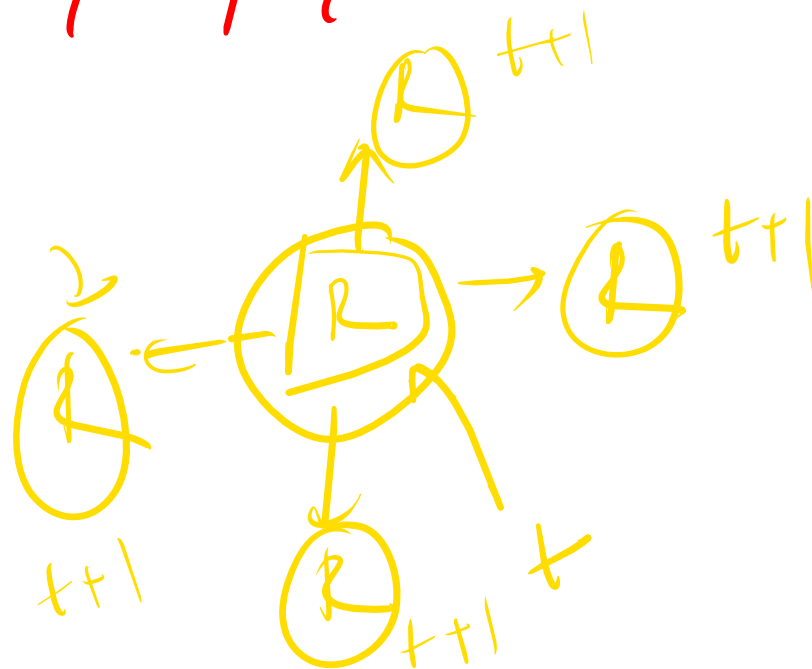
ans  
②

min no.  
of mins.

|   |   |   |   |   |
|---|---|---|---|---|
|   | 0 | 1 | ② | 1 |
| 1 |   | ② |   | 0 |
|   |   |   |   | 1 |
|   |   |   | ② |   |

|  |  |   |   |   |   |   |
|--|--|---|---|---|---|---|
|  |  | 2 | 1 | 1 | 2 | 1 |
|  |  | 2 | 1 | 1 | 2 | 1 |
|  |  | 2 | 1 | 1 | 2 | 1 |
|  |  |   |   |   | 2 | 1 |
|  |  |   |   |   | 1 | 2 |
|  |  |   |   |   | 1 | 2 |

$\leftarrow \begin{pmatrix} i \\ d \end{pmatrix} \begin{pmatrix} j \\ t \end{pmatrix} \rightarrow$



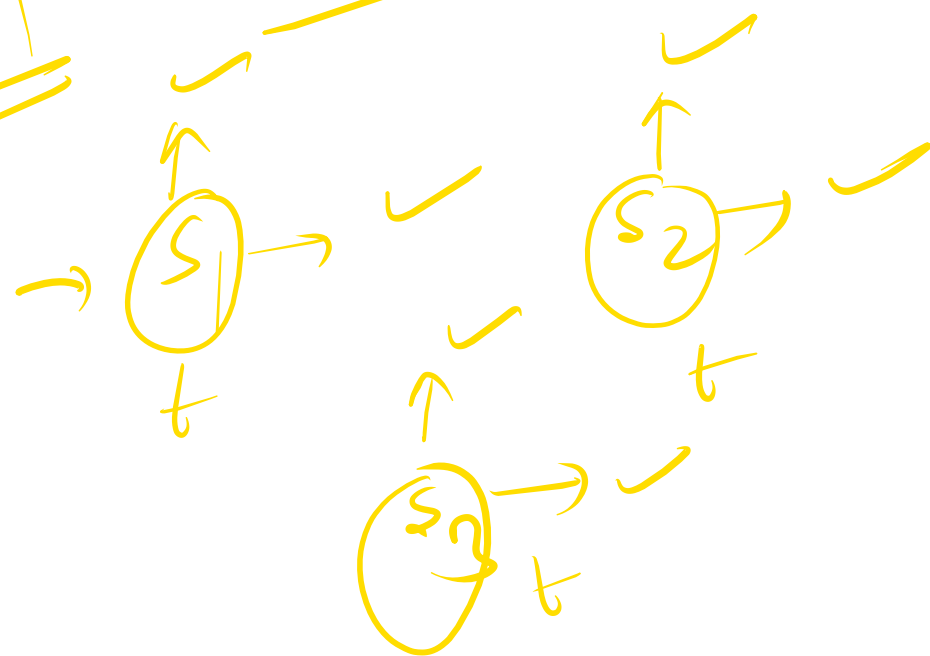


①

time

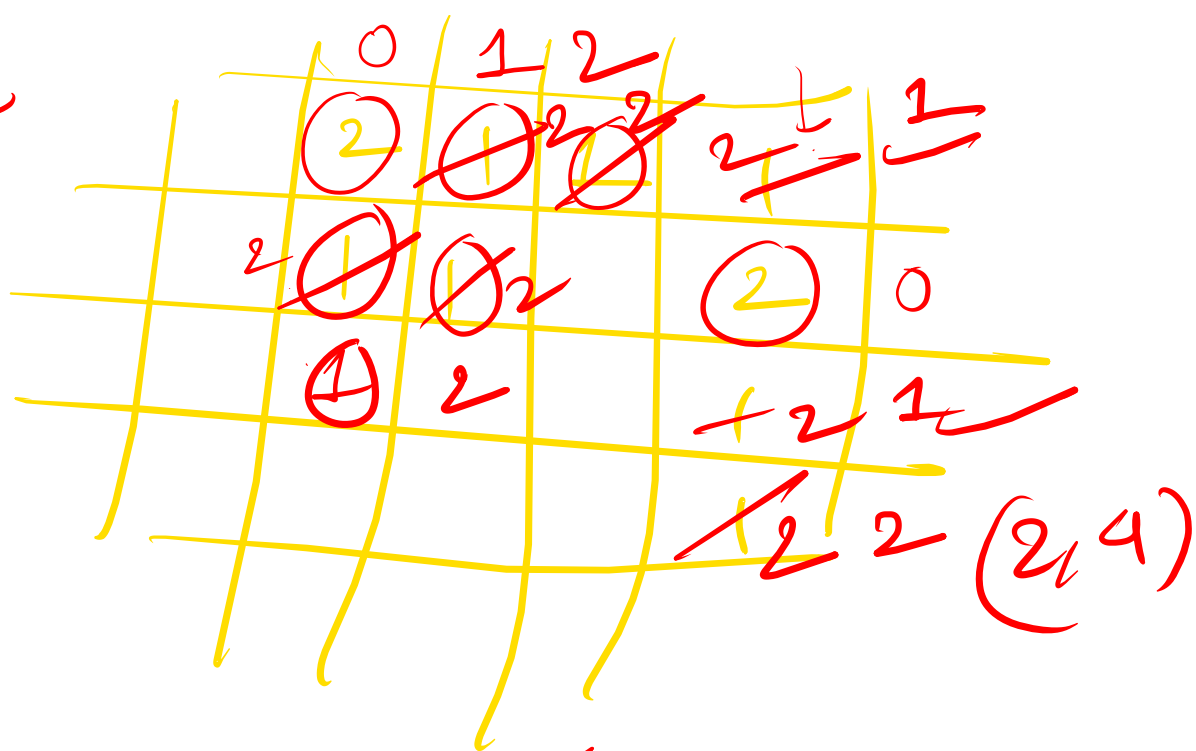
↓  
minimised

t+1



BFS  
 $(s_1 \dots s_n)$

2



q: ~~(0,1)~~ ~~(1,4)~~ ~~(0,2)~~ ~~(1,1)~~

(0,4)

(1,1)

(0,1)

(1,4)

(0,2)

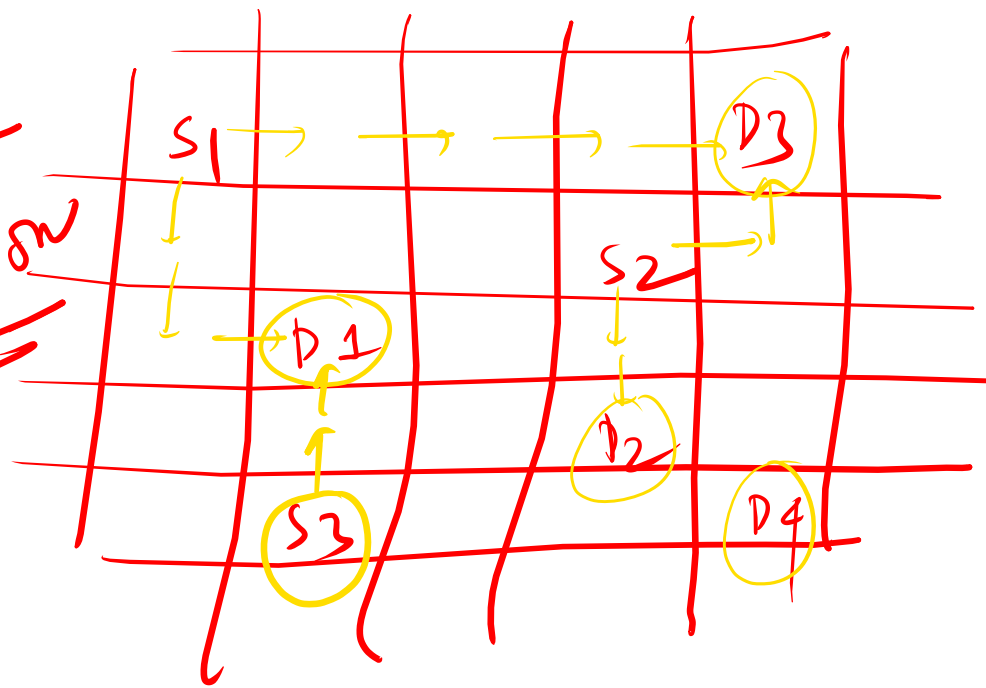
~~(0,4)~~ ~~(2,4)~~

~~(0,3)~~ ~~(1,2)~~

(3,4)



$s_i$   $\rightarrow$  distance  
 to  
 nearest  
 destination



$s_1 \rightarrow 3$   
 $s_2 = 2$   
 $s_3 = 2$

$O(K * (n * m))$

$s_i \rightarrow \text{BFS}$   
 $(d_j)$

there  
 $s_2 \quad s_3 \quad \dots \quad \underline{s_n}$   
 min ✓

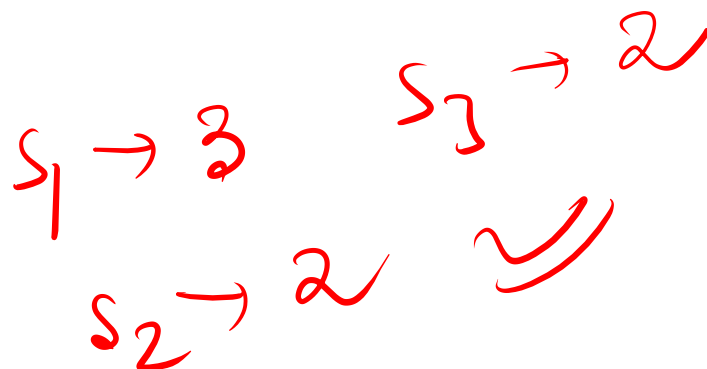
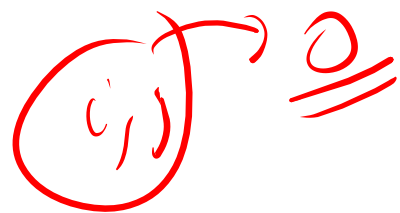
|   |   |   |   |   |
|---|---|---|---|---|
| 3 | 2 | 2 | 1 | 0 |
| 2 | 1 | 2 | 2 | 1 |
| 1 | 0 | 1 | 1 | 2 |
| ✓ | 1 | 1 | 0 | 1 |
| 3 | 2 | 2 | 1 | 0 |

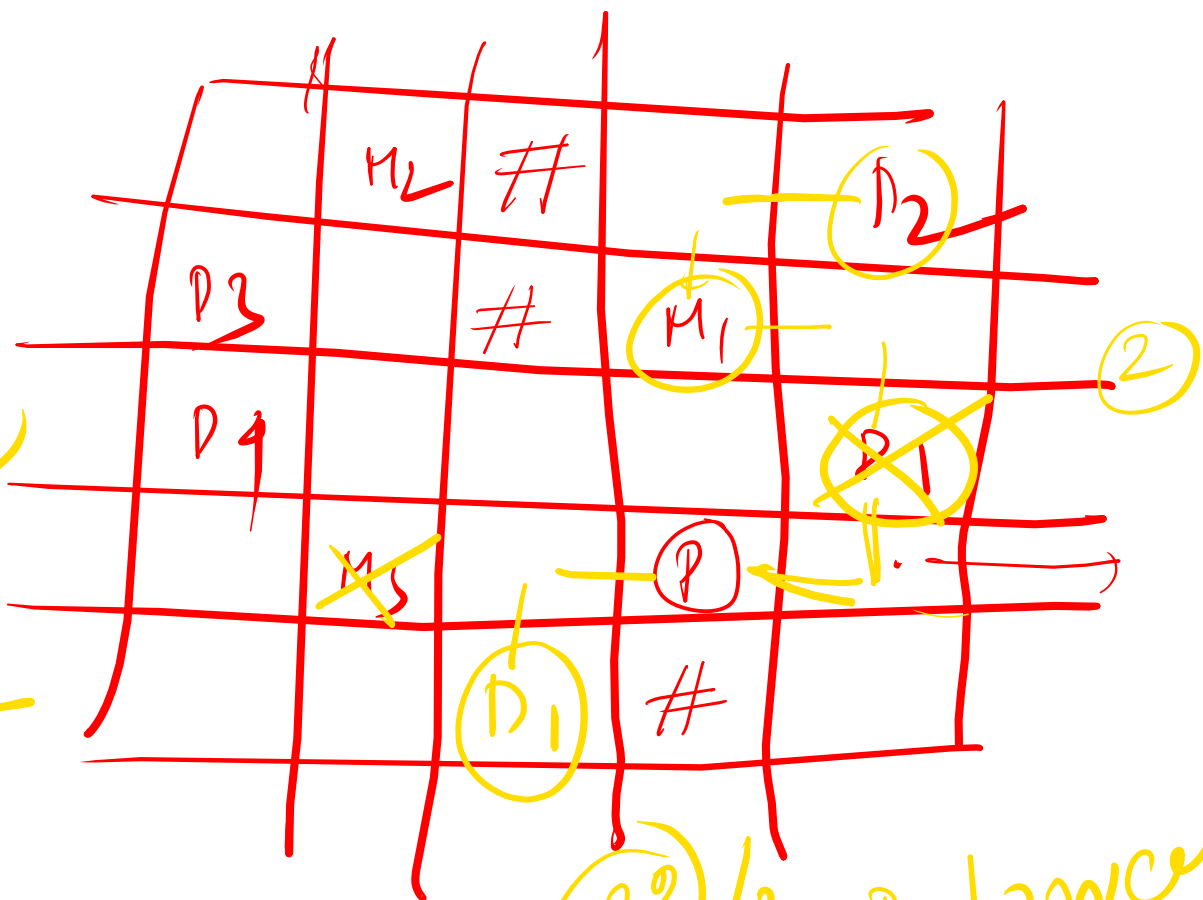
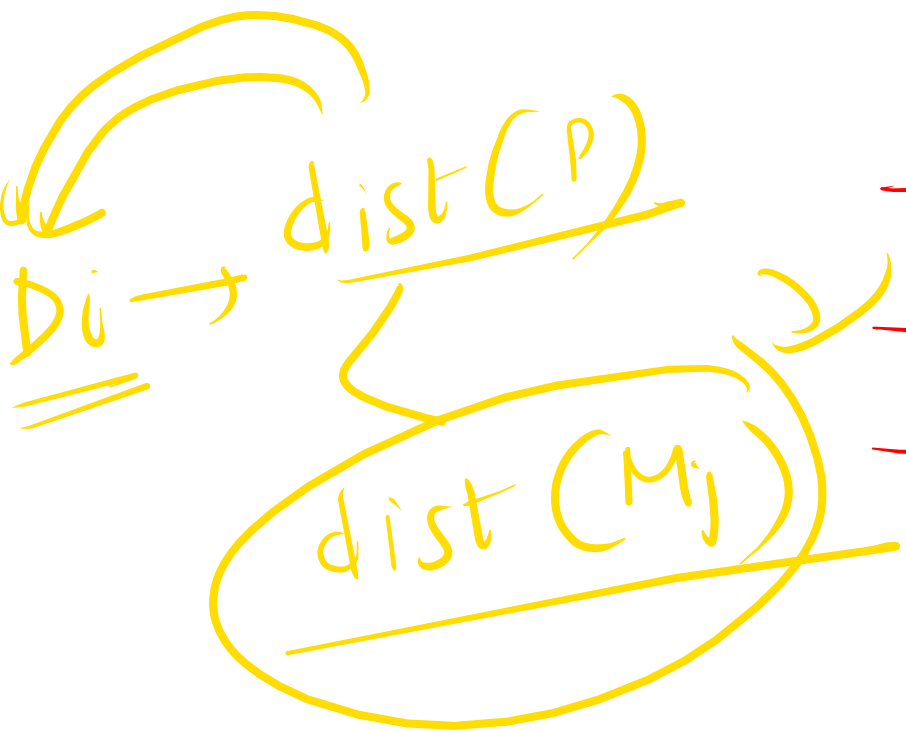
$n * m$

distance

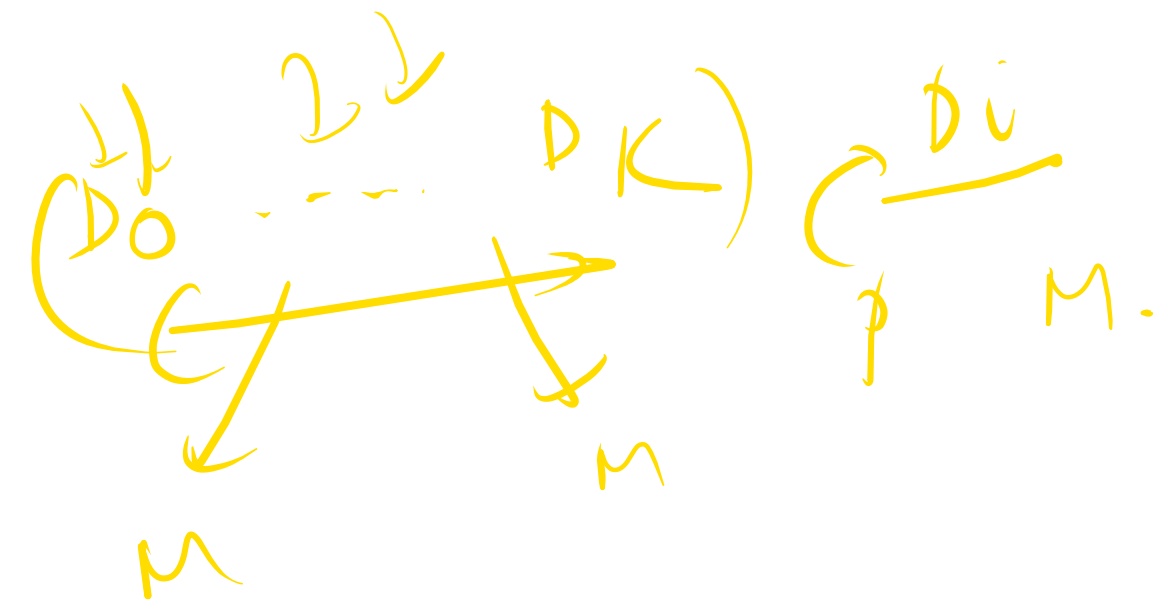
|   |   |   |   |   |
|---|---|---|---|---|
| 2 | 1 | 1 | 1 | 3 |
| ✓ | ✓ | ✓ | ✓ | ✓ |
| ✓ | ✓ | ✓ | ✓ | ✓ |
| ✓ | ✓ | ✓ | ✓ | ✓ |
| ✓ | ✓ | ✓ | ✓ | ✓ |

time/distance





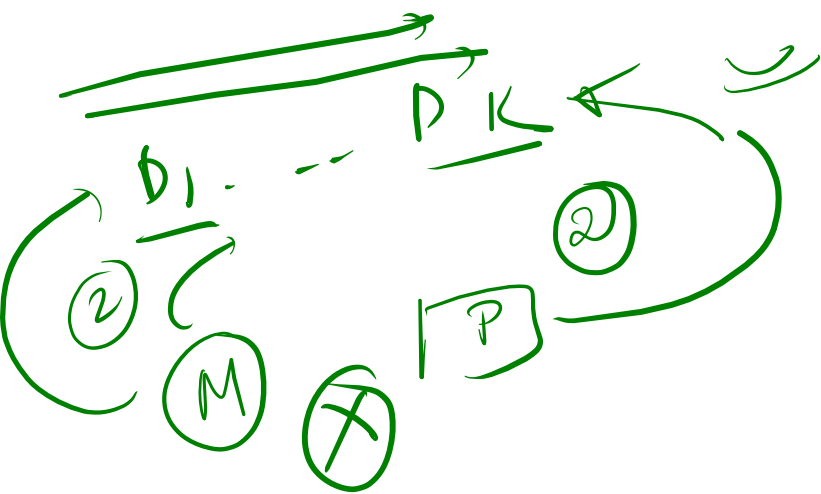
$K \times N \times M$



$\frac{(2^9)}{2}$  ? lower distance

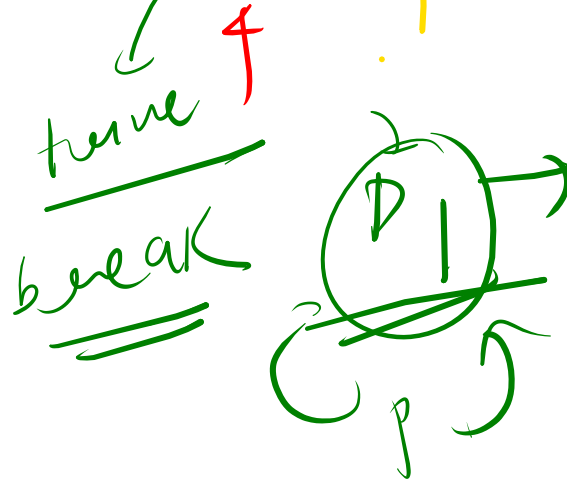
$\underline{D_i}$



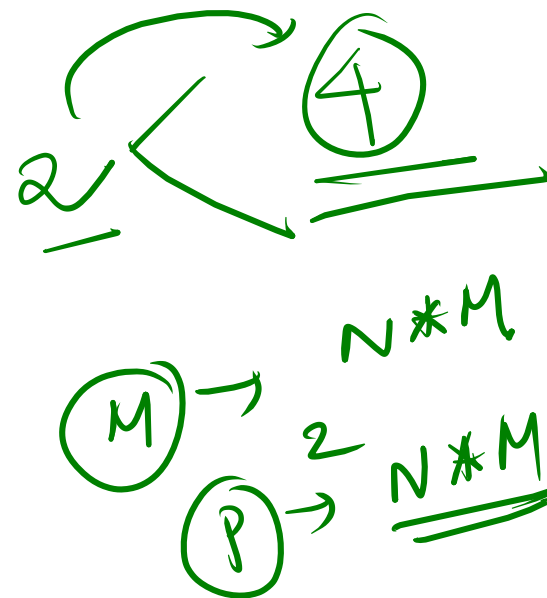


$D_0 \rightarrow ?$   
 $D_0 \rightarrow M$   
 $O(K * N * M)$

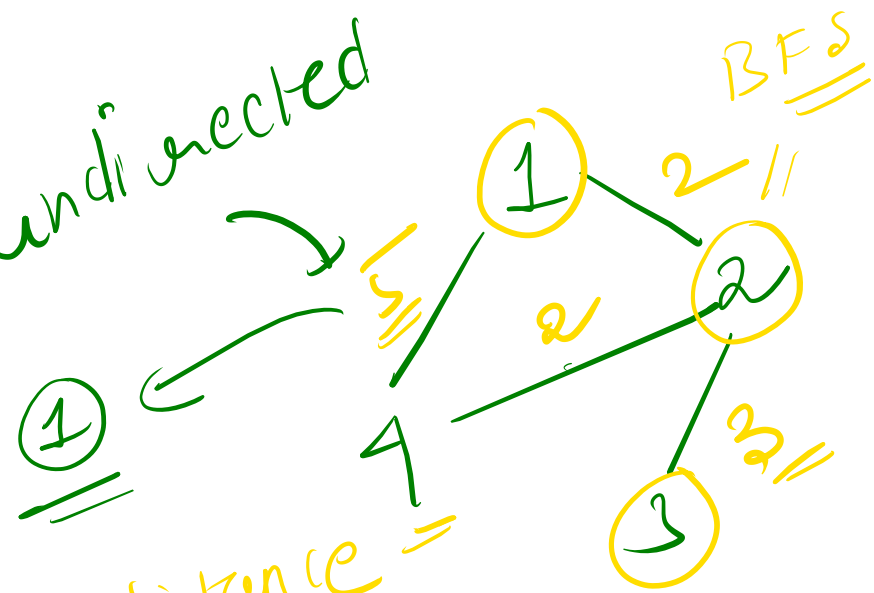
|   |       |       |         |          |       |
|---|-------|-------|---------|----------|-------|
|   | 1     | $M_2$ | #       | 1        | $P_2$ |
| 2 | $P_3$ | 1     | #       | $M_1$    | 1     |
| 3 | $P_4$ | 2     | 2       | 1/1      |       |
|   |       | 3     | 3/1     | <u>P</u> | 1     |
|   |       |       | $P_1/2$ | #        |       |



$M_1 \dots M_n$



undirected



cost  $\rightarrow$  distance  $=$   
 $u-v$



weighted

1 : 0  
2 : 2  
3 : 5

4 : 5 / 4

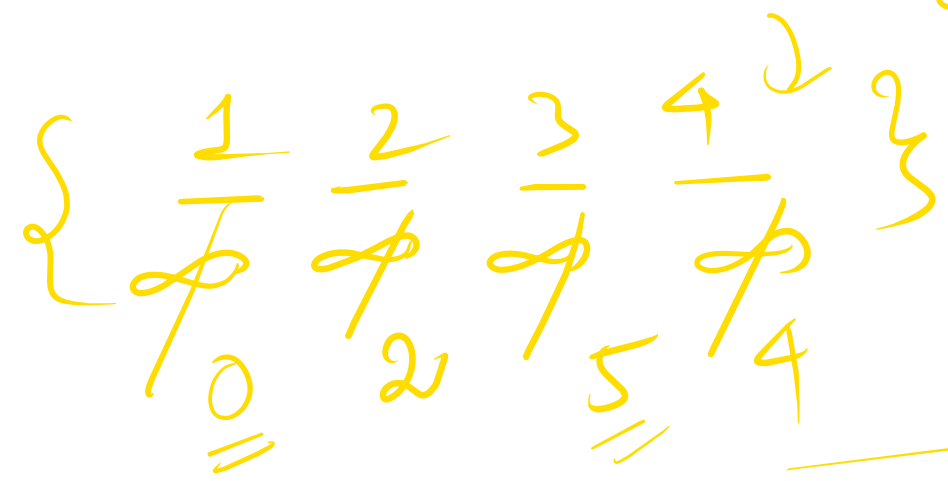
4 : 4<sup>2</sup>

BFS

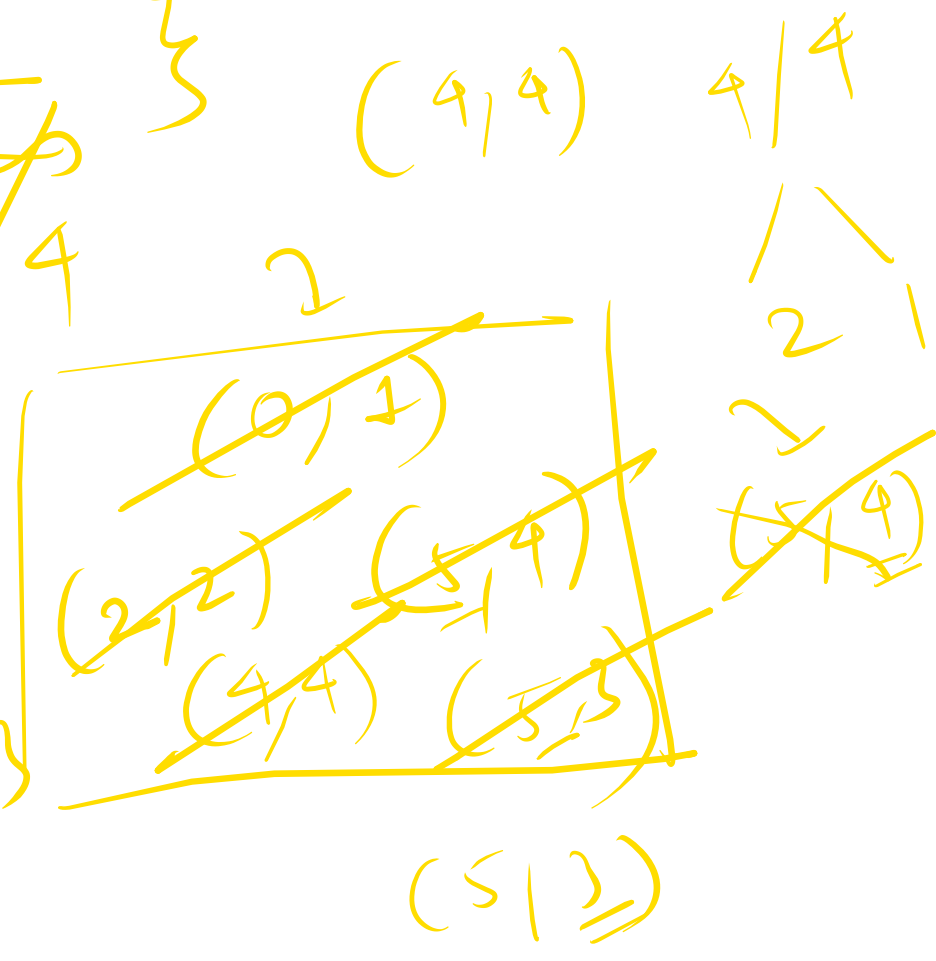
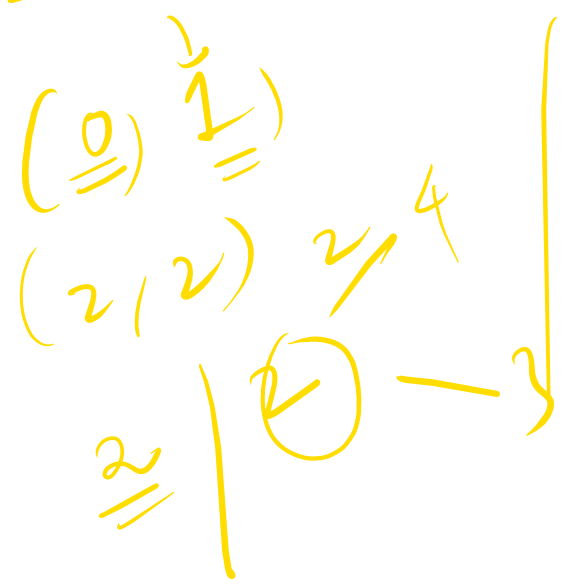
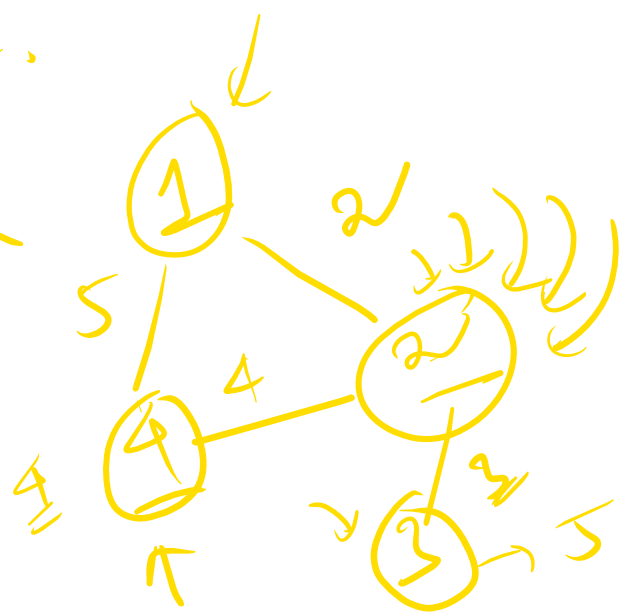


# Dijkstra

$\infty \rightarrow \text{INT\_MAX}$



implement.  $\leftarrow$



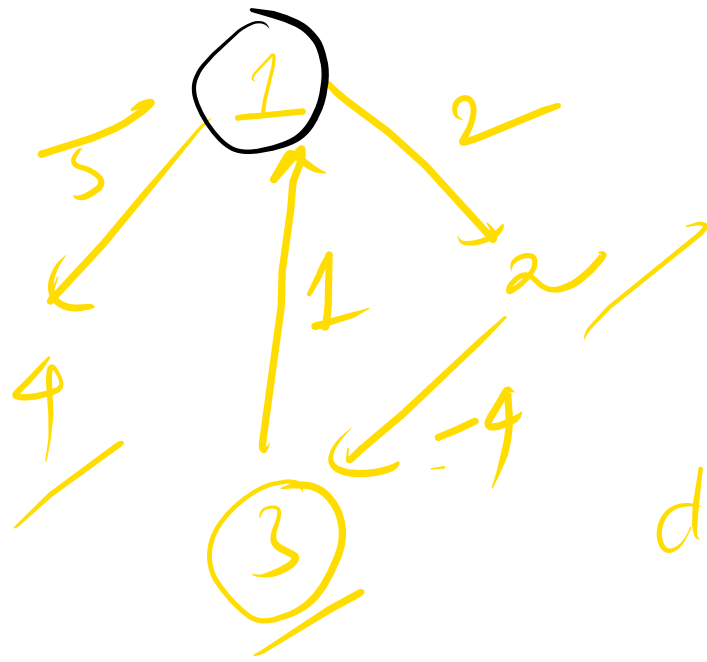


-ve edge

↓  
directed graph



|                          |
|--------------------------|
| <del>(0, 1)</del>        |
| <del>(2, 2)</del> (3, 4) |
| <del>(-2, 3)</del>       |



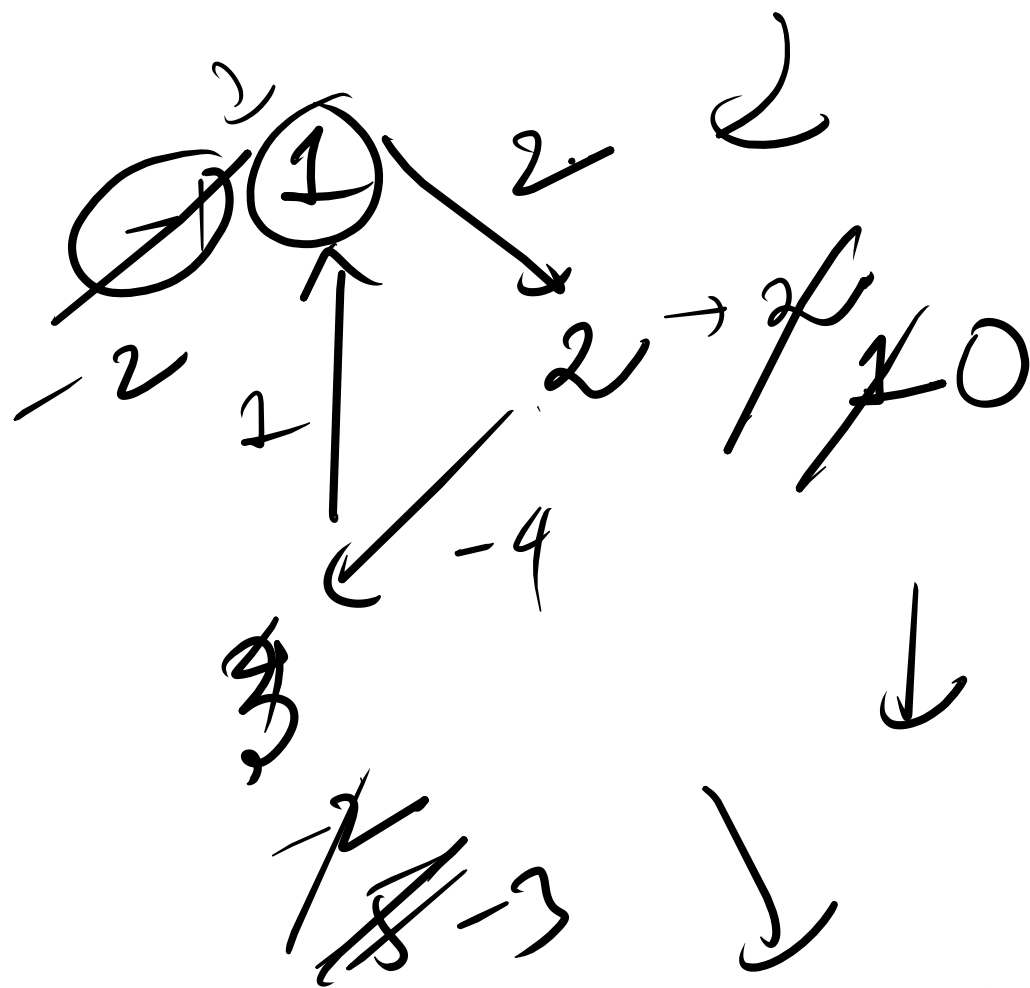
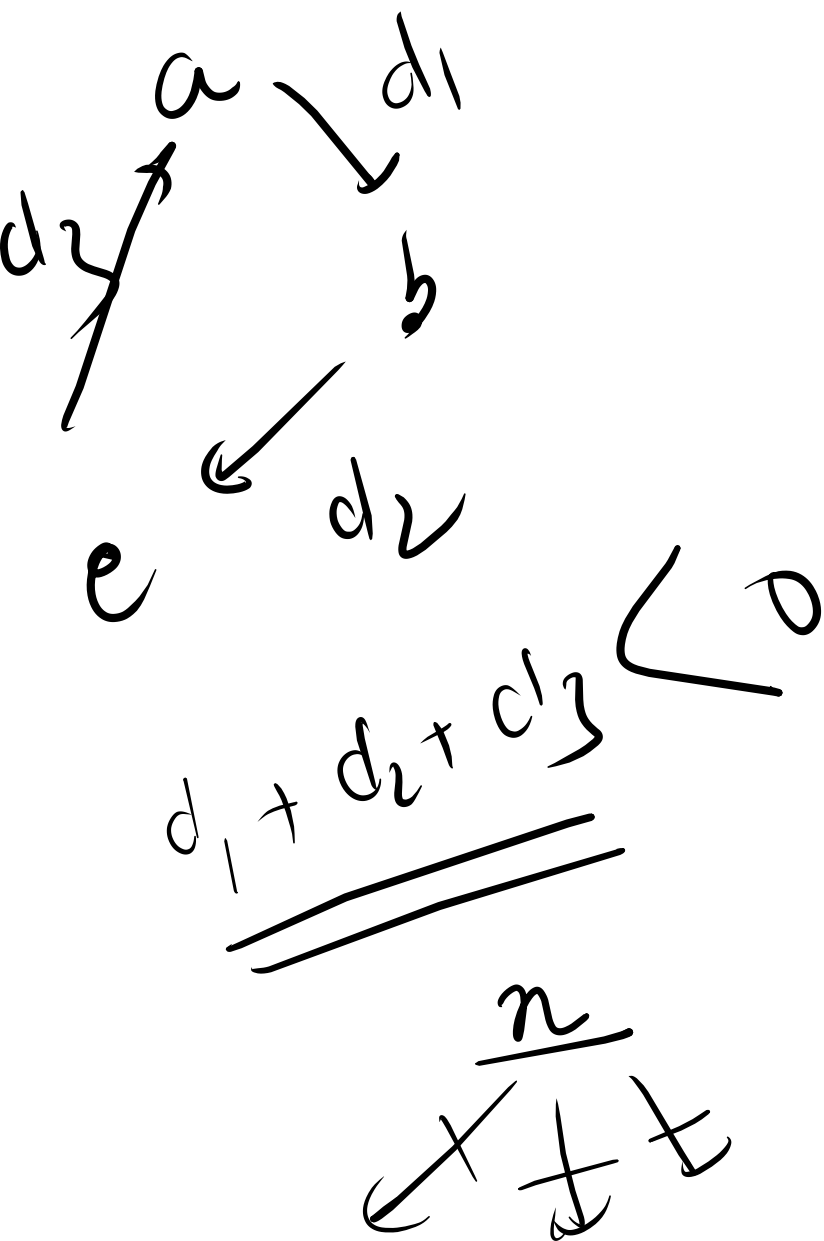
min dist

④ →

dijkstra

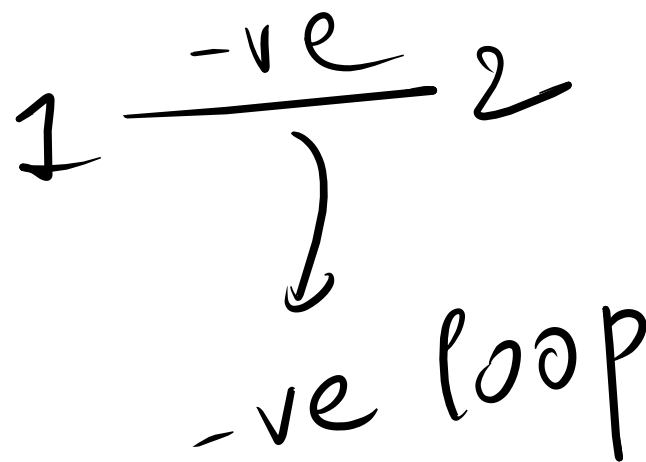
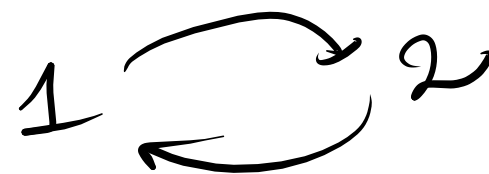
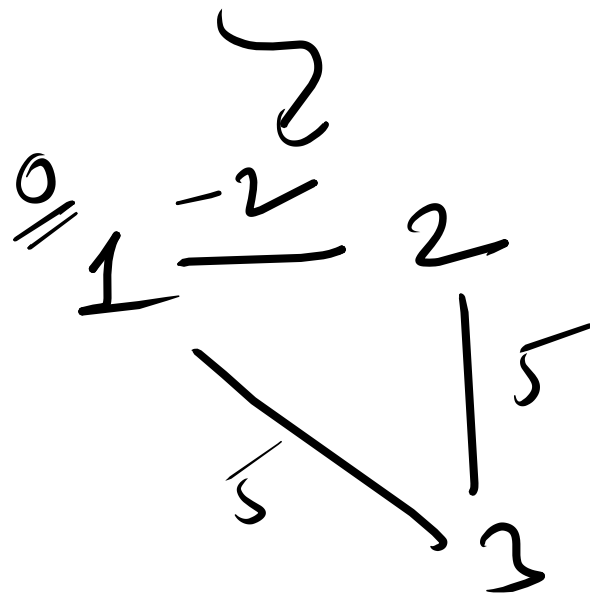
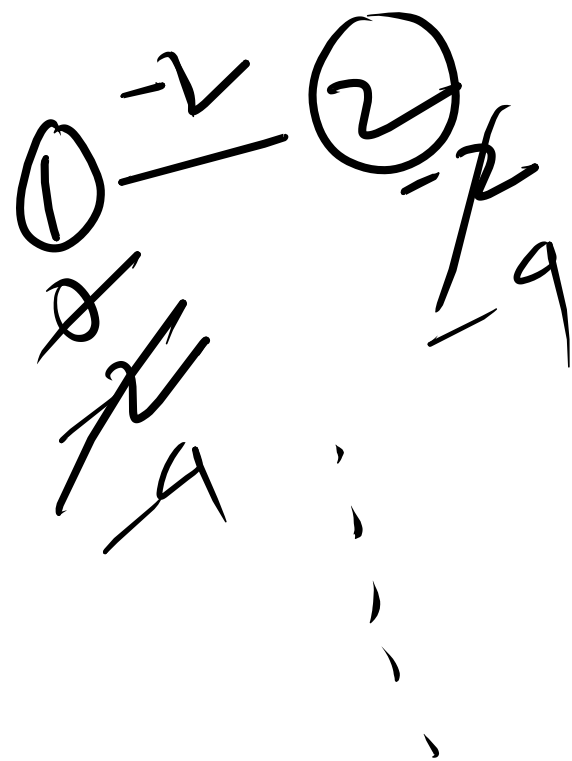
1 → 0 / -  
2 → 2  
3 → -2

3 → 1 → ④?  
-2 + 1



edge relaxation { Bellman  
 ford

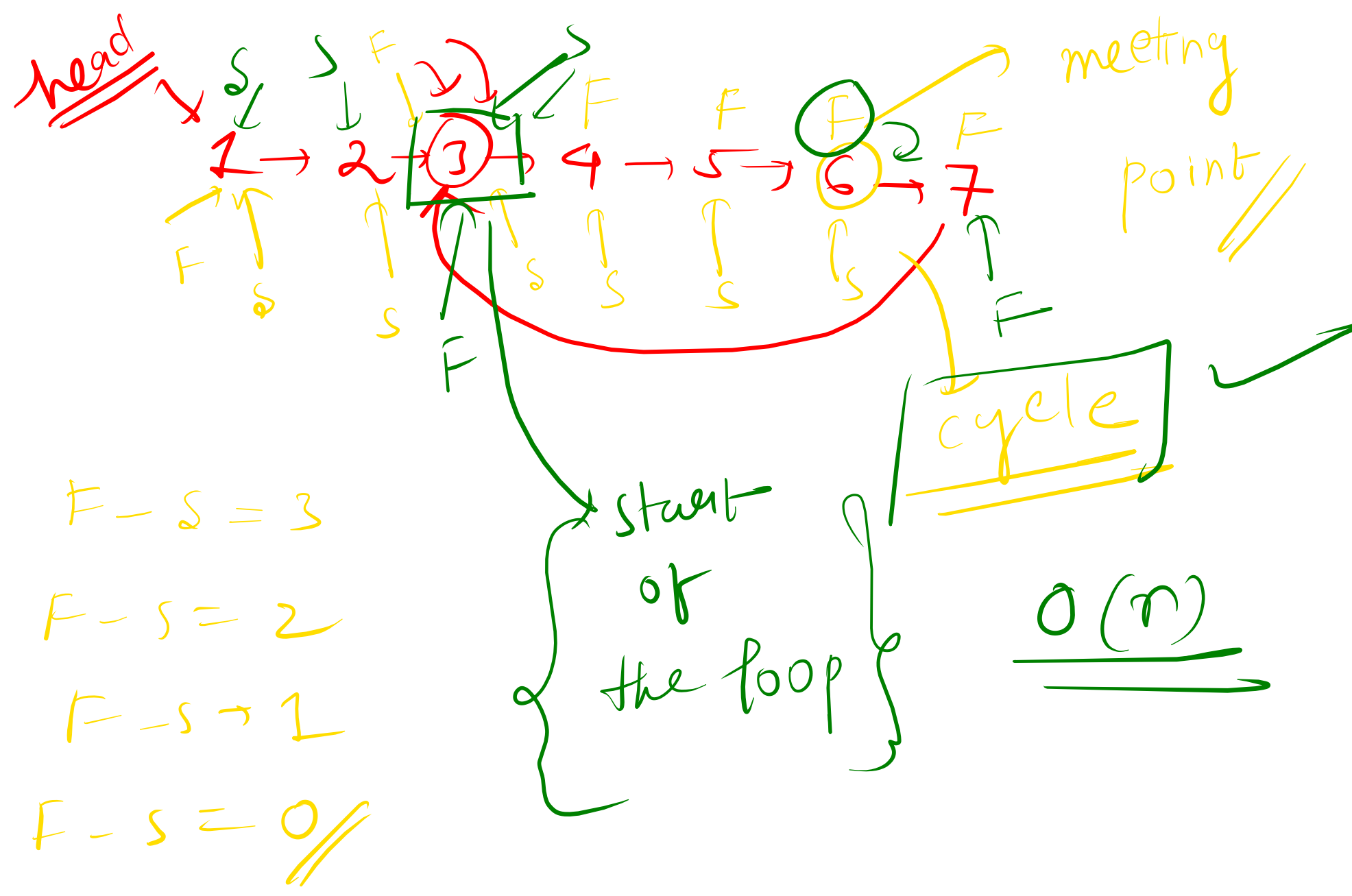
-ve



loop  
S - F → 1 step

F → 2  
S → 1

n — prev  
↓  
length of  
loop



F - S = 3  
F - S = 2  
F - S = 1  
F - S = 0 //

start  
of  
the loop

O(n)

Time

1 → true

2 → true

3 → true

4 → true

5 → true

6 → true

x

$O(n)$

sc

x

$h_1 \rightarrow$  ①

2

3

④

5

6

⑦

⑧

$h_2 \leftarrow$

$O(n+m)$

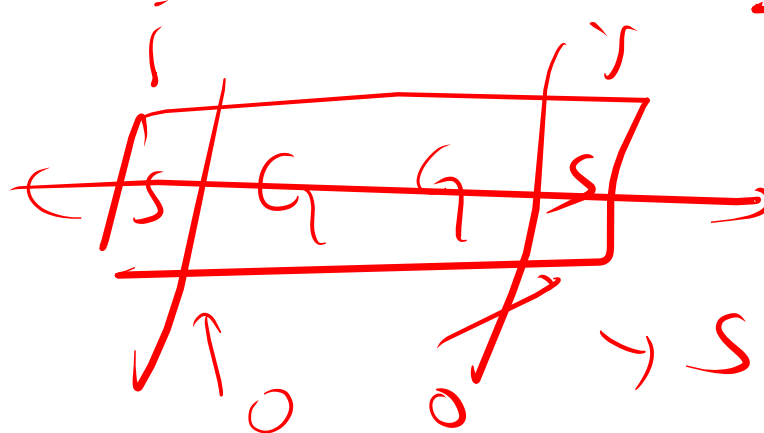
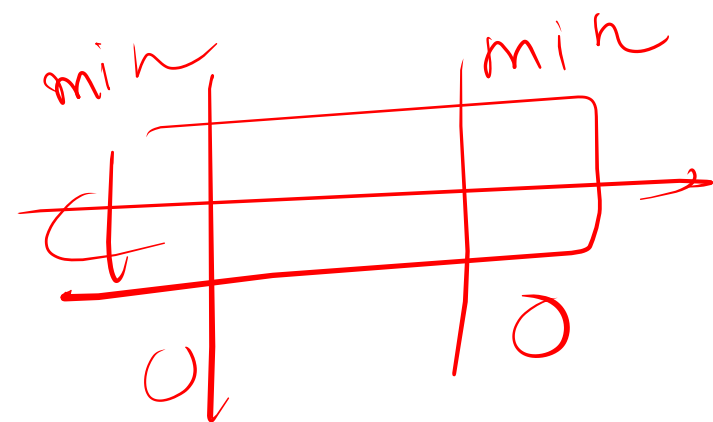
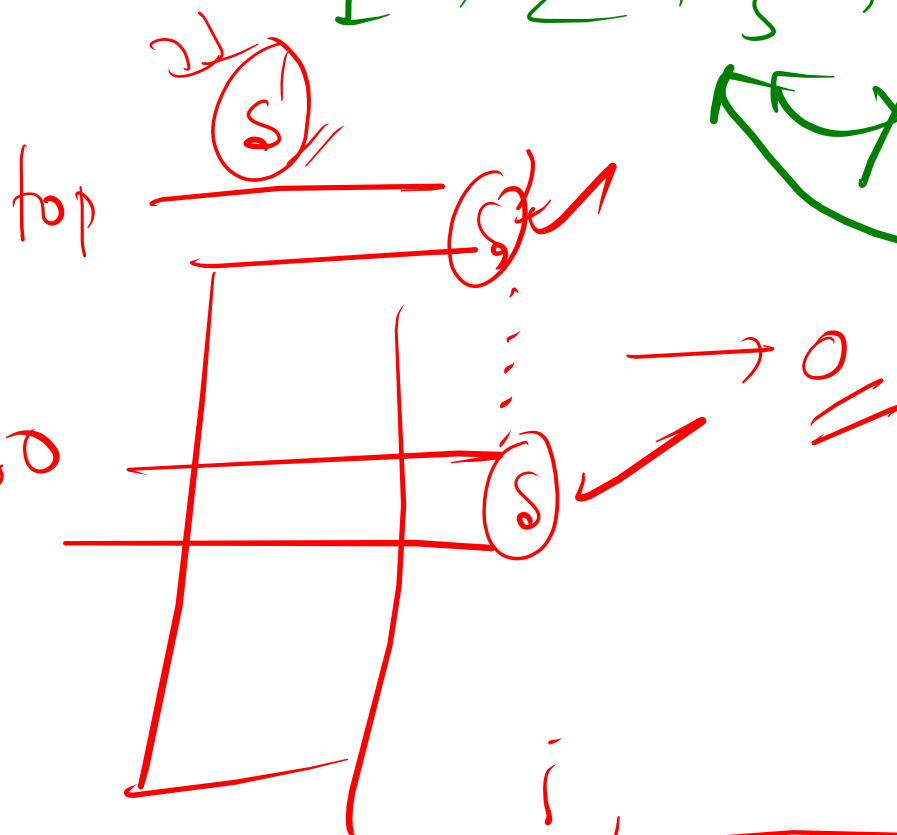
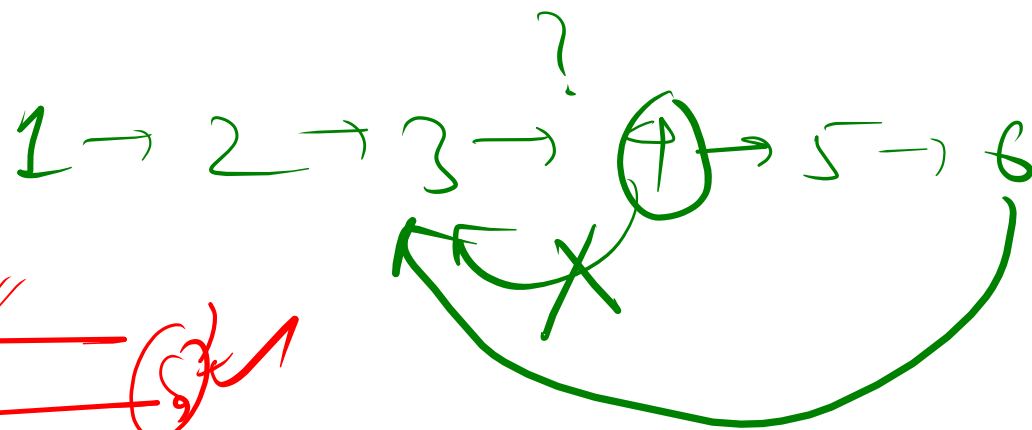
return

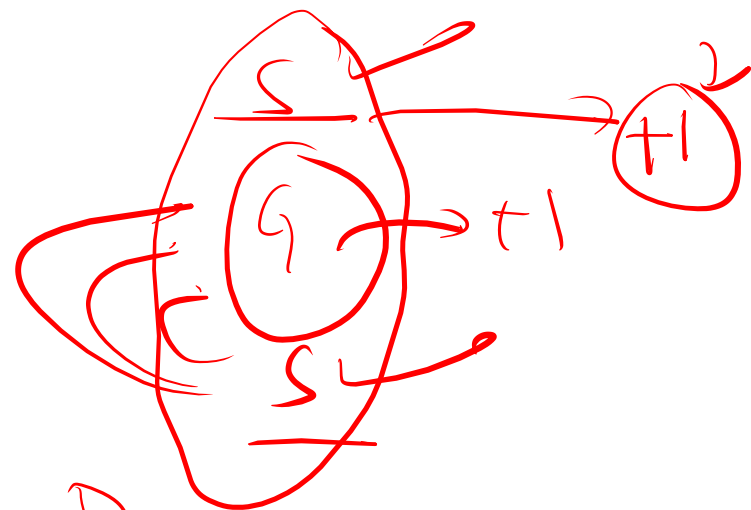
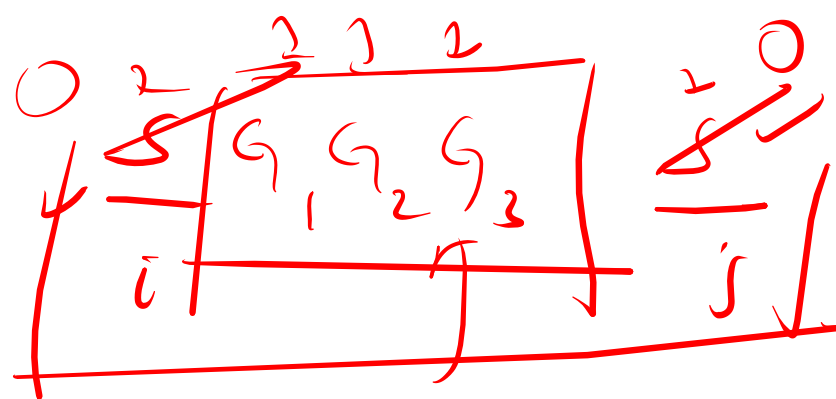
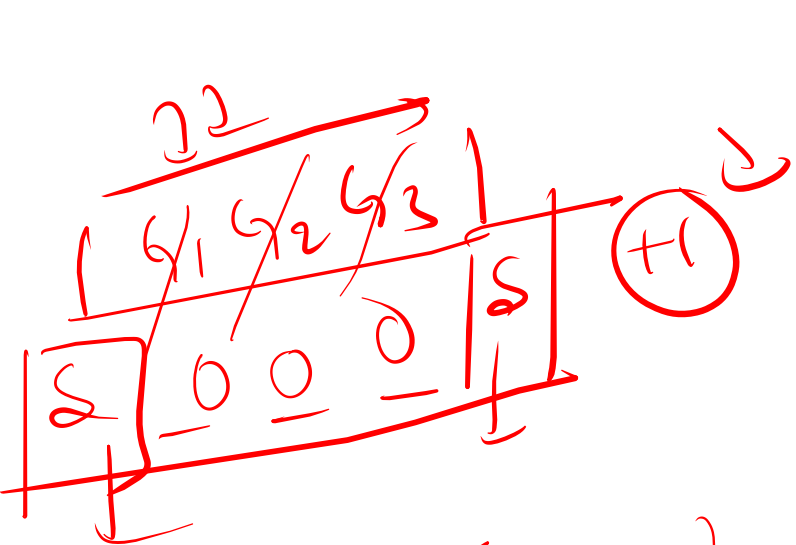
$$\begin{aligned} & n+m \\ & n+m \\ & O(2n+2m) \\ & \underline{\underline{O(n+m)}} \end{aligned}$$

$$\begin{aligned} & \underline{\underline{c_1 / c_2}} \\ & \text{const.} \\ & O(1) \rightarrow \underline{\underline{sc}} \\ & \text{add } \textcircled{4} \\ & \text{return } \underline{\underline{n}} \end{aligned}$$

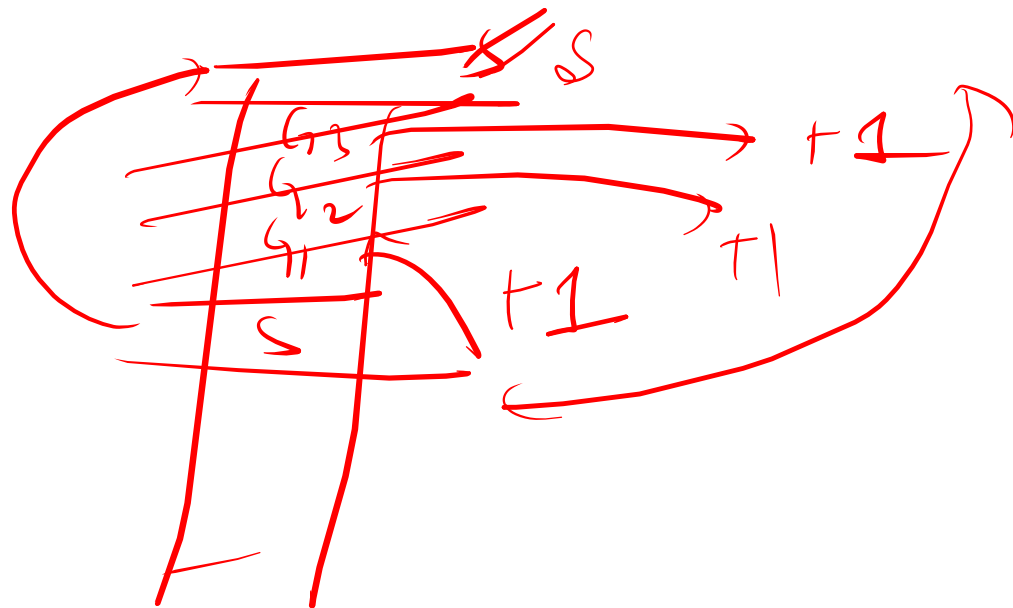
$$\begin{aligned} L_1 &= 6 \\ L_2 &= 5 \end{aligned}$$



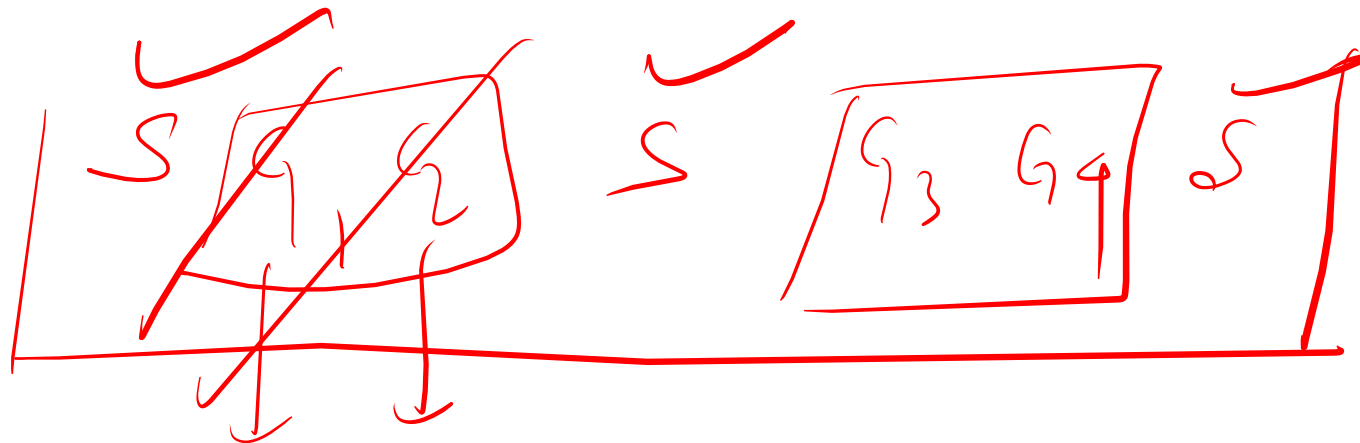




$$g_1 < g_2 < g_3$$







0 (D)

H1 H2 S  
~~G4 → H1~~  
~~G3 → H2~~  
 S  
 S

S  
 S  
 S → H1

+S

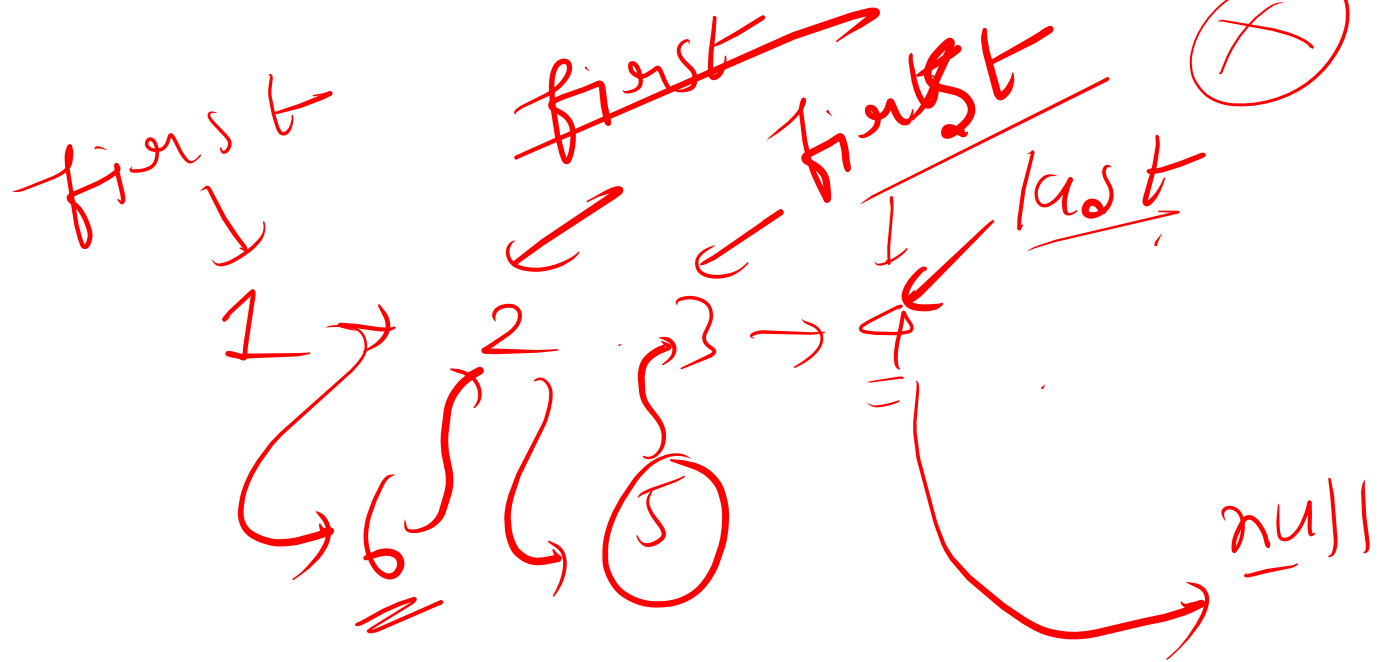
1 → 2 → 3 → 4 → 5



1 → 5 → 2 → 4 → 3 } null

{  
1 → null  
2 → 1  
3 → 2  
4 → 3  
5 → 4  
}

{ 1 → 2 → 3 → 4 → 5 → 6 }



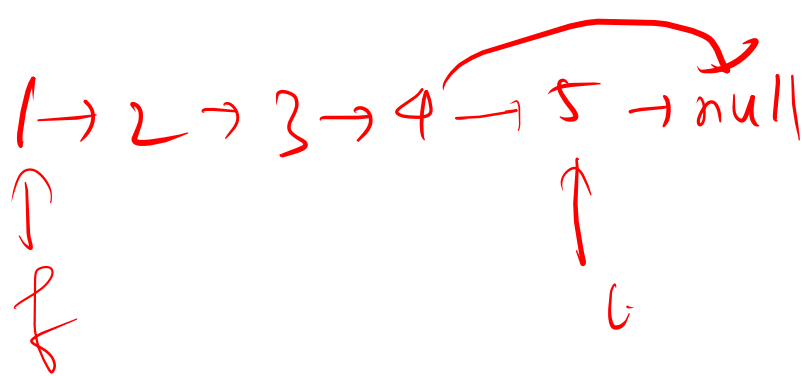
$\{$   
 $O(n)$   $\rightarrow$   
 $-$

HM

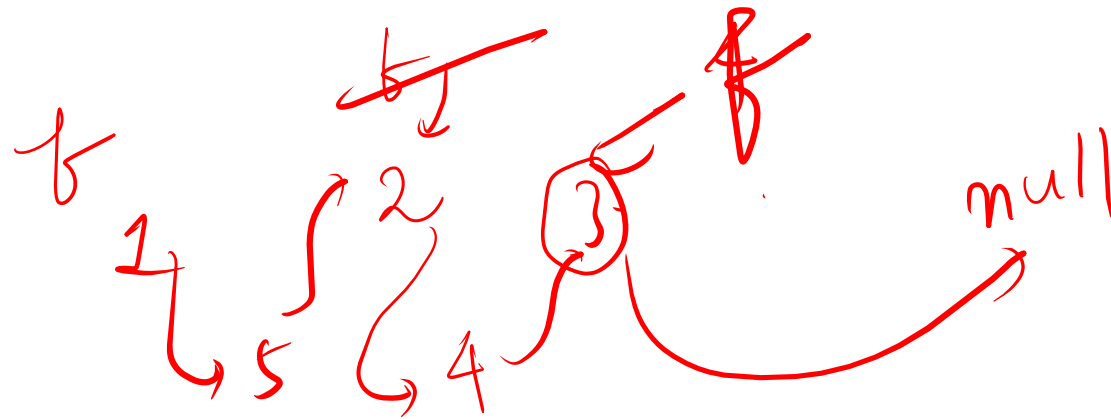
$1 \rightarrow \text{null}$   
 $2 \rightarrow 1$   
 $3 \rightarrow 2$   
 $4 \rightarrow 3$   
 $5 \rightarrow 4$

$\}$

H/S



$\{ \mid = \underline{\underline{\text{mid}}}$



$(1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6)$

$(7 \rightarrow 8 \rightarrow 9 \rightarrow 10)$   
 $(11 \rightarrow 12)$

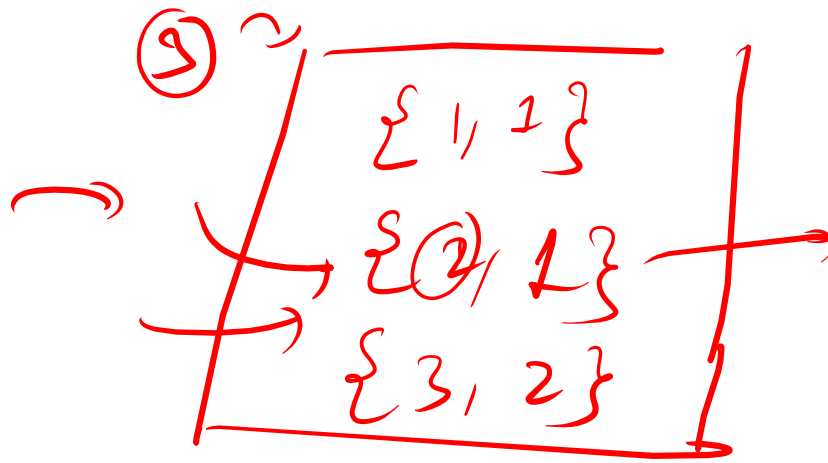
$[7 \rightarrow 8 \rightarrow 11 \rightarrow 12 \rightarrow 9 \rightarrow 10]$

$(1 \rightarrow 2 \rightarrow 3 \rightarrow 7 \rightarrow 8 \rightarrow 11 \rightarrow 12 \rightarrow 9 \rightarrow 10 \rightarrow 4 \rightarrow 5 \rightarrow 6)$

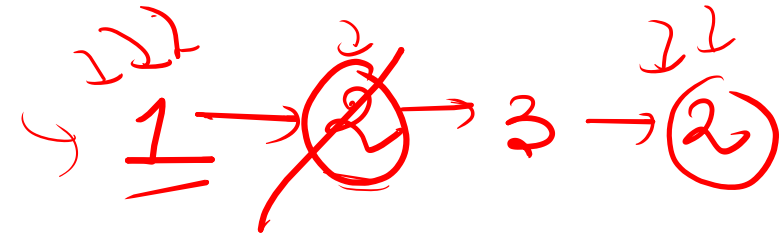
# Least recently used

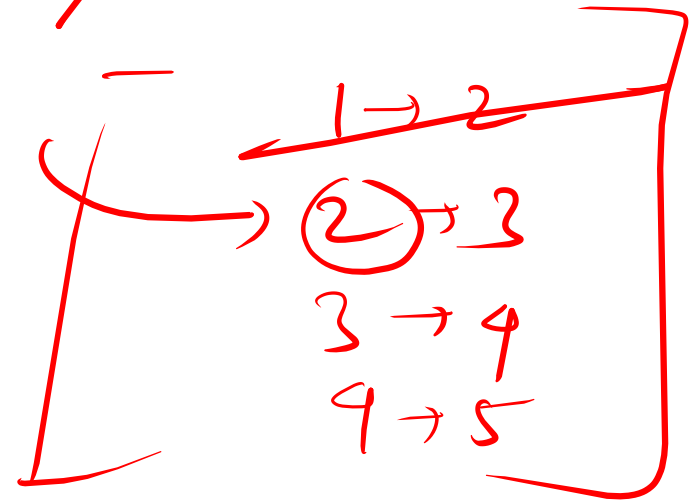
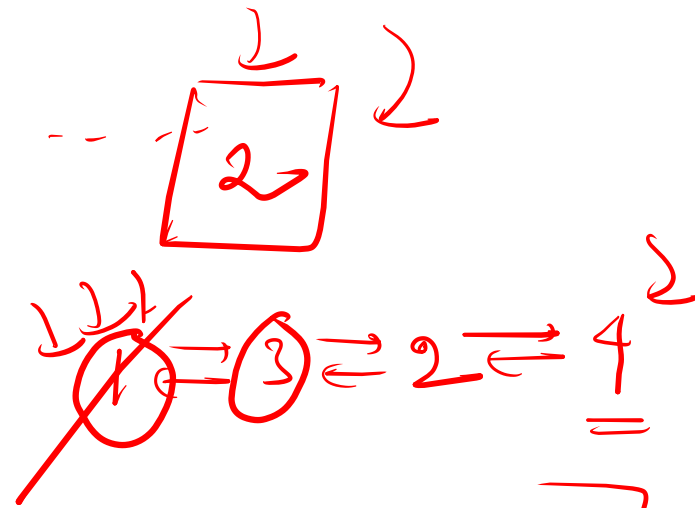
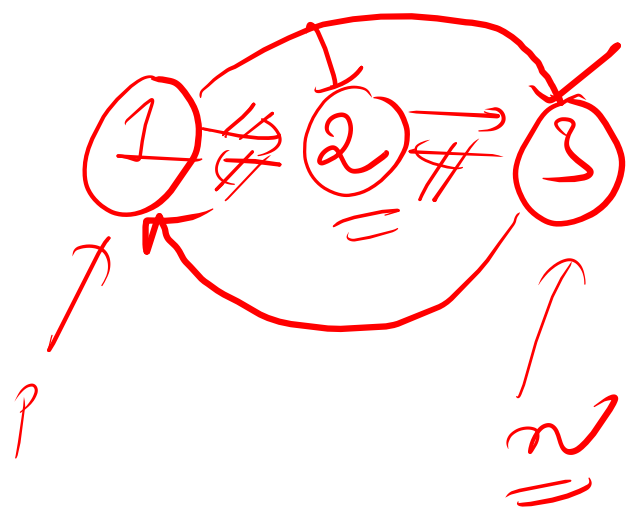
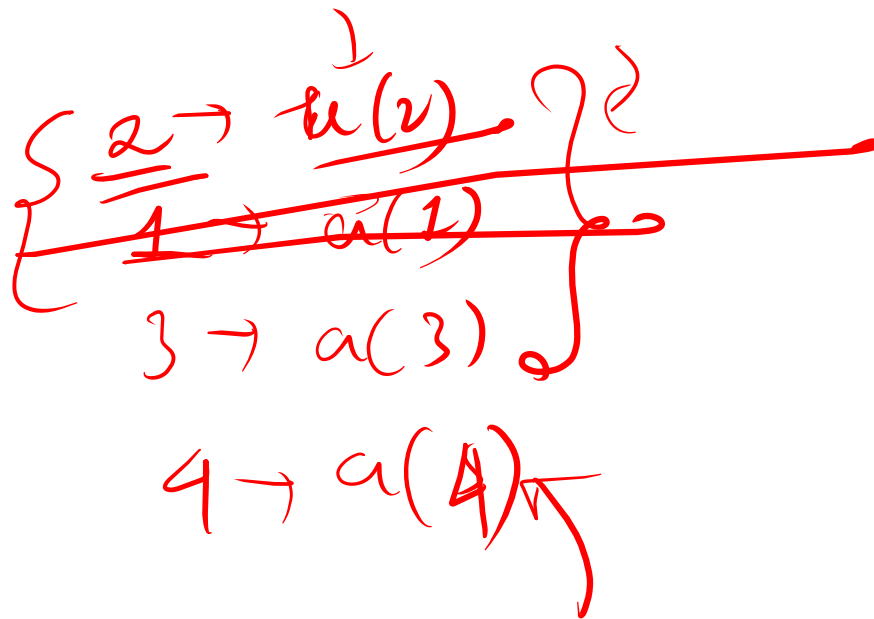
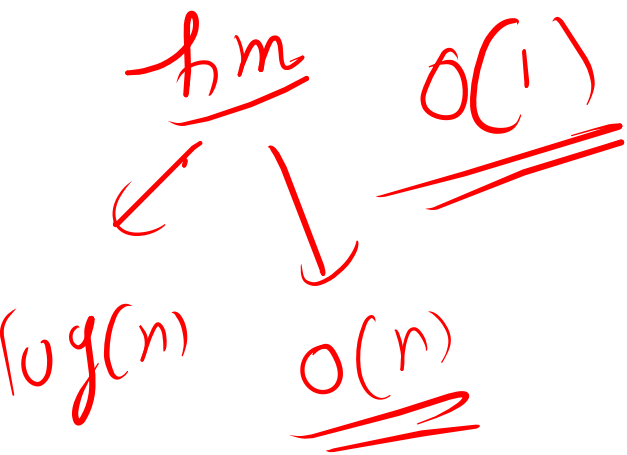
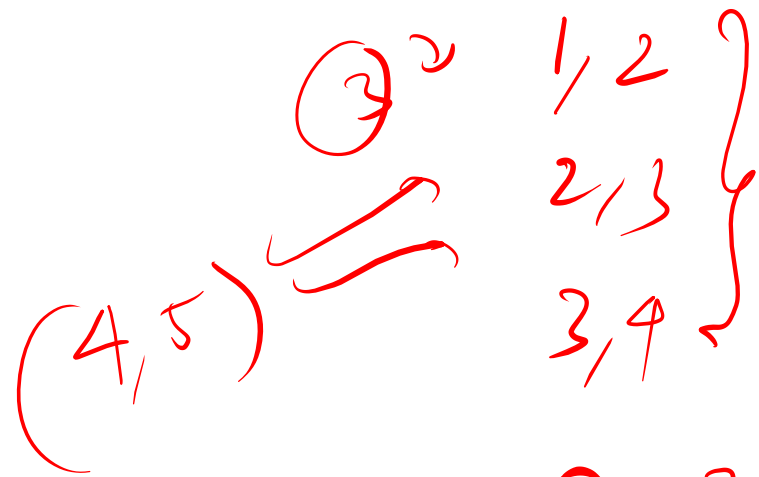
map

h<sub>1</sub>  
add  
→ h<sub>2</sub>  
LL →  
✓



1)



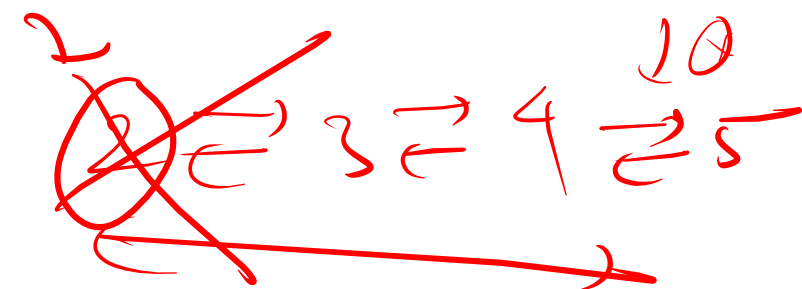
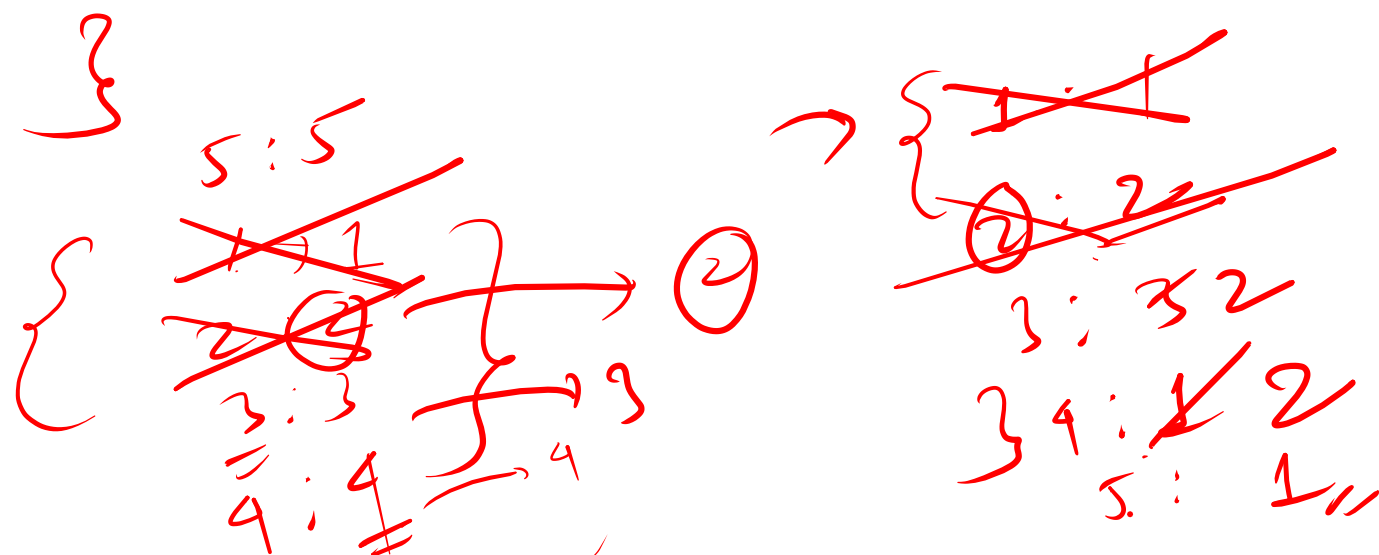
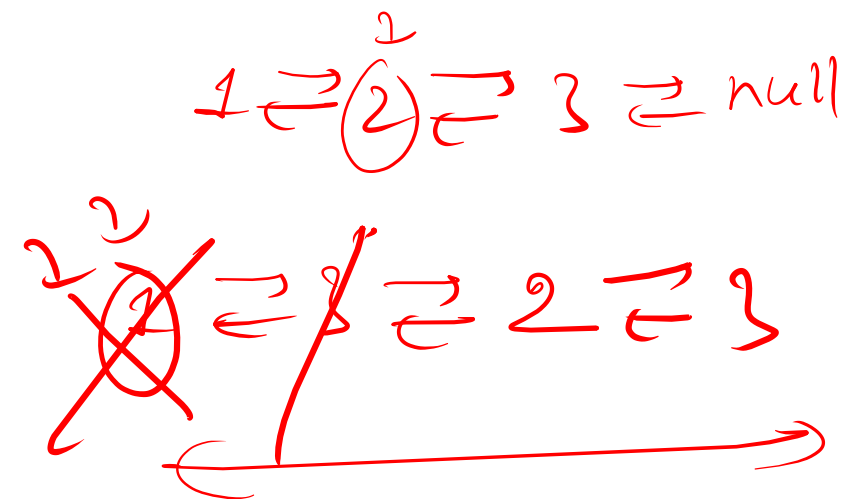
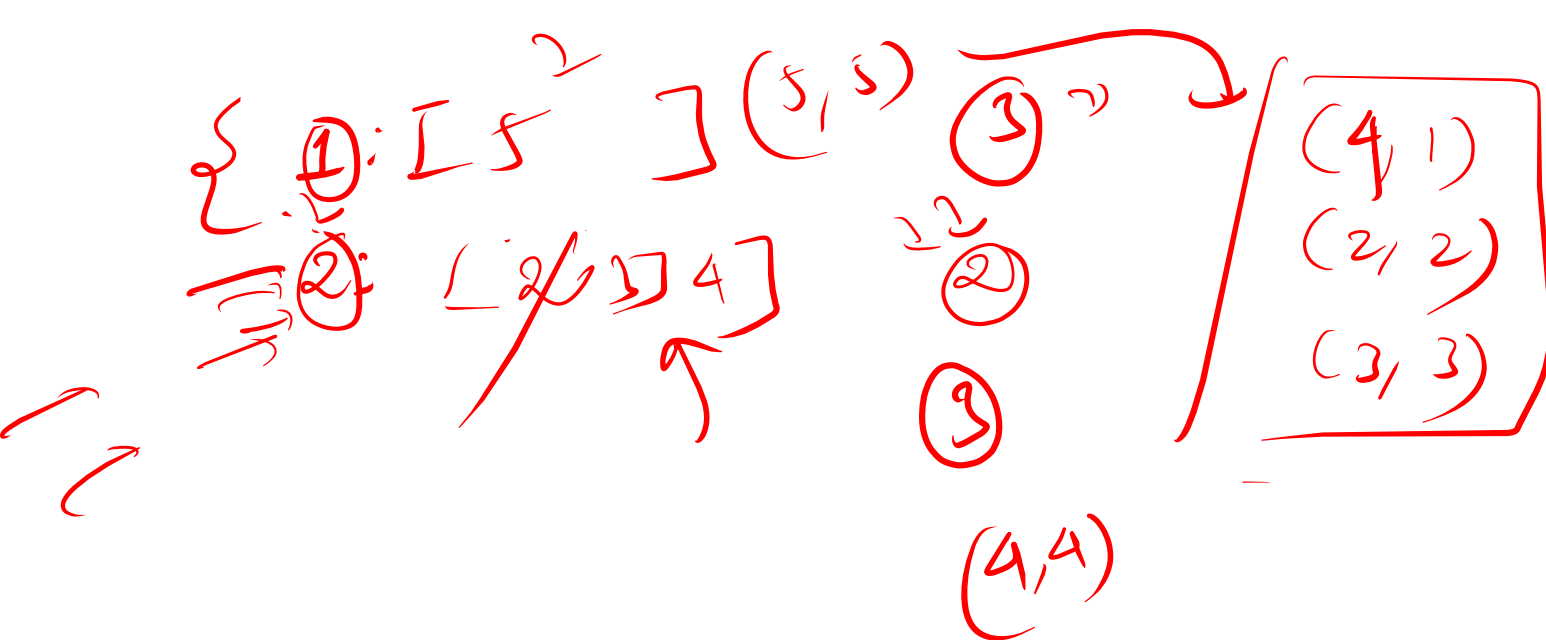


hLL

⇒ list → STL

· push\_back

· push\_front



code